Battles and Spoils: Evidence from the British Museum

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Research Agenda

I study innovation, institutions, and organizations.

- **Innovation in emerging markets**

  – how China's rise spurs entrepreneurship in other emerging markets, especially where Chinese business models suit local conditions (with Lerner, Moscona, and Yang; *Journal of Political Economy*, accepted)

  – what explains India's low patenting and how a reform that expands institutional capacity improves innovation outcomes (with Kala and Lerner)

- **Political economy of organizations**

  – sanctions' spillovers on scientific collaboration (with Fenske, Haseeb, and Kala)

  – how firms respond to political risk in colonial India (with Fenske and Gupta); how proponents respond to costs and voting incentives (with Pi and Wang)

  – the role of spoils of war in military compensation (today's talk)

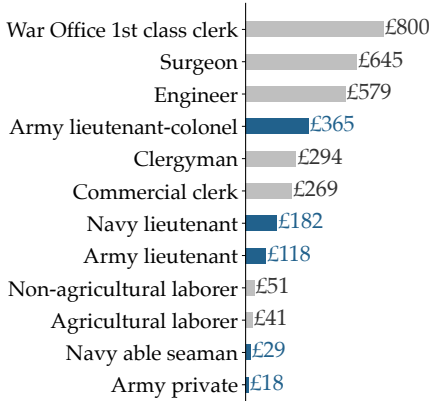
Organizational Design and Compensation

- **Organizational design.** Organizations pursue collective objectives through people with their own interests. Rules, authority, and rewards shape their participation (Joseph and Sengul, 2025).
- **Returns beyond salary.** Ownership, revenue shares, and career opportunities can also reward participation (Gibbons, 2005; Silverman and Ingram, 2017).
- **Imperial warfare.** Existing literature typically studies imperial warfare as a choice of the state (Gallagher and Robinson, 1953; Acemoglu et al., 2012; Hoffman, 2015). Economic models of conflict emphasize the importance of material rewards in motivating agents (Blattman and Miguel, 2010; Ray and Esteban, 2017).

Motivation: Low Military Pay in the British Empire

Military historians generally describe British officers as poorly paid, despite the dangers of imperial service.

- Army officers earned under half the pay of civil service clerks of equivalent grade (Spiers, 1980).
- War Secretary Windham, 1806: "none could enter the army as officers, who had not something else besides their pay" (Bruce, 1980).
- MP Elphinstone, 1873: "impossible for [Navy] officers … to live upon their pay".



Pounds per year, c. 1871. Sources: Spiers (1980); Beckett (2018); *Navy List* (1870); Williamson (1982).

Motivation: Spoils of War As Compensation

One explanation is that the spoils of war, an enduring feature of warfare, formed a component of compensation.

- Eleven documented prize funds from the Indian Mutiny (1857–58), financed by auctioning looted treasure, totaled about £1.2 million (US$260 million today). Banda and Kirwee alone paid a lieutenant roughly three years' pay and Commander-in-Chief Campbell £22,200 (US$4.9 million today).
- Frederick Engels, writing on the Indian Mutiny: "some of these officers have made, literally, their fortunes", "officers and soldiers went into the town poor and debt-ridden, and came out suddenly enriched".

However, spoils are hard to trace and measure, and systematic evidence on how much battles yielded and who benefited remains limited.

Motivation: Spoils in the British Museum

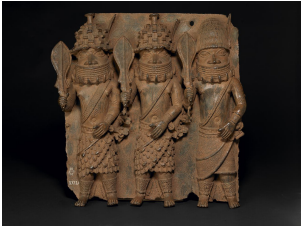
- Founded in 1753, the British Museum built one of the world's largest collections, with an aim to collect cultural treasures across the globe.
- Its catalog records their origins, acquisition dates, and sources.



Chalice
Magdala, Ethiopia, 1868



Admonitions Scroll
Beijing, China, 1900



Plaque
Benin City, Nigeria, 1897

Research Questions

To show the role and impact of spoils, I study three research questions:

1. Did battles systematically yield spoils?
2. Did the officers who fought spoil-rich battles benefit?
3. Did spoil-rich battles affect officers' career choices?

This Paper: Setting

I study these questions in the British Empire in the nineteenth century.

- Britain held the largest empire and fought hundreds of battles that varied in timing and location.
- The state authorized the taking of spoils and helped organize their sale and distribution as prize money (Spiers, 2020; Aldous, 2020).
- Records of artifacts, battles, officers, and their wealth survive.
- Officers also paid a large sum (up to 12 years' annual pay) for ranks in the army (Bruce, 1980), providing a setting to study officers' career choices.

This Paper: Empirical Design

1. Did battles systematically yield spoils?

   I aggregate British Museum acquisitions and British battles to city or grid level as a panel, and use event studies with city, year, and country by year fixed effects to trace Museum acquisitions around British battles.

2. Did officers benefit from spoil-rich battles?

   I compare wealth at death among battle commanders of similar rank and background, sorting battles into high-spoils and low-spoils by the Museum inflows that followed them.

3. Did spoil-rich battles affect career choices?

   I use a difference-in-differences design to compare commission purchases in regiments that participated in the Indian Mutiny with noncombat regiments.

This Paper: New Data

- **Artifacts**
  - Scrape 1.5 million British Museum online collection records and geocode each artifact.
  - Digitize 900,279 pages of auction catalogues and identify 1,803 campaign-related lots.
- **Battles**
  - Digitize the *Dictionary of Battles and Sieges* (8,401 entries), identify 474 battles that Britain fought, and manually check and verify each entry.
- **Commanders**
  - Research the biographies of 317 named commanders of British battles by hand.
  - Digitize 16.1 million probate records from 837,358 pages to match commanders and military officers to their wealth at death.
- **Prize and rank purchasing**
  - Digitize 656,357 pages of the *London Gazette* to extract prize fund notices and all military promotions, purchases, exchanges, and retirements.

Preview of Results

1. British battles systematically generated spoils.
   - Next-year British Museum acquisitions from battle cities rise 8.6-fold (from 0.7 to about 6 artifacts) and stay higher for decades.
   - Battle-induced arrivals within three years make up a fifth of acquisitions from those cities.
   - Artifacts taken from 17 of 19 campaigns still appear in auction catalogues after 10 years.

2. Commanders of spoil-rich battles benefited.
   - £13,400–14,400 more wealth at death per artifact-heavy battle (US$3.0–3.3 million today).
   - Prize money could explain on average 40% of wealth at death where recorded.

3. Spoil-rich battles affected officers' career choices.
   - After the spoil-rich Indian Mutiny, purchases in participating regiments rose by 45%.
   - Outsiders exchanged into or bought first commissions in those regiments.

Related Literature

- **State capacity: motivating agents** (Besley and Persson, 2009; Finan et al., 2017; Khan et al., 2019; Johnson and Koyama, 2014; Hillmann and Gathmann, 2011)
  - The largest nineteenth-century empire still paid its officers with a share of spoils.
- **Economics of conflict: who fights and who benefits** (Blattman and Miguel, 2010; Ray and Esteban, 2017; Fenske and Voth, 2024; Dube and Vargas, 2013; Sánchez de la Sierra, 2020; Allen, 1998)
  - What state officers gained from battles: support for purchase and prize money as an incentive contract.
- **Costs and returns of empire** (Acemoglu et al., 2001; Nunn, 2008; Dell, 2010; Stahn, 2023; Hobson, 1902; Davis and Huttenback, 1986; Offer, 1993; Xu, 2018)
  - The taking of cultural objects in war, and private returns to the officers who carried it out.
- **Economics of art, museology, and collecting** (Ashenfelter and Graddy, 2003; Oosterlinck, 2017; Fisman and Wei, 2009; MacKenzie, 2009; Jasanoff, 2005; Hicks, 2020; Sarr and Savoy, 2018)
  - Quantitative evidence on the taking over a century, beyond famous individual cases.

Broader Relevance

How organizations allocate rewards shapes individual behavior (Holmström and Milgrom, 1991), and related cases persist today:

- Police asset forfeiture: U.S. police departments in some states can retain proceeds from seized assets (Baicker and Jacobson, 2007).
- Tax farming redux: revenue-linked bonuses for tax collectors in Pakistan (Khan et al., 2016).
- More broadly, similar questions arise when startups offer equity alongside low current pay, or when public officials supplement their salaries through bribes.

This paper traces the returns military officers received and how spoils shaped entry and advancement within the army.

Roadmap

1. **Background**
2. Data
3. Artifacts
4. Officers
5. Conclusion

From Battle to Museum

Founded in 1753, the British Museum was the first national, public, and secular museum in the world. Yet, many of its holdings have a contested provenance. The British Museum Act 1963 prohibits the Trustees from disposing of objects.

▷ can the Museum return objects?

Spoils could reach the British Museum by various routes.

1. **Prize auctions:** prize agents auctioned captured property to dealers and collectors, from whom the Museum could later buy.
2. **Private holdings:** military personnel could hold on to some spoils and sell or give them to the Museum.
3. **Government transfers:** the Crown or government could reserve objects and later transfer some to the Museum.

The Institution and Legality of Loot

- **Army: prize auctions, shares by rank.** The Crown granted captured property to the troops, and prize agents auctioned the loot (Spiers, 2020). ▷ army prize
  - Wolseley's *Soldier's Pocket-Book*, a field manual for officers, set out the procedure, since "no orders on the subject are to be found in our regulations" (Wolseley, 1886).

- **Navy: Prize Acts, shares by rank.** From 1708, an Act of Parliament granted captured property to the officers and men (Aldous, 2020). ▷ navy prize
  - Prize "was one of the great inducements in the naval service, especially to officers" (Lavery, 1989).

- **"Loot":** British troops in India adopted the Hindustani *lūṭ*, plunder, as their own word for what they captured. ▷ the word "loot"

- Condemned in the Lords (1861) and by Prime Minister Gladstone (1871), but never banned until the Hague Conventions of 1899 and 1907.

Military Pay and Commission Purchase

- Military historians generally agree that Britain paid its officers and men badly (Barnett, 1970; Lavery, 1989; Chandler and Beckett, 1994). ▷ pay in context
  - Army officers' pay ranged from £96 a year for an ensign to £365 for a lieutenant-colonel and was fixed from 1806 to 1914 (Spiers, 1980).
  - After 1815, some 3,000 of the navy's 4,000 lieutenants sat on half-pay of about £90 a year (Lewis, 1960).

- From 1660 to 1871, purchase was the ordinary route to commission and promotion up to lieutenant-colonel (Bruce, 1980). ▷ the tariff and premiums ▷ how purchase worked ▷ credit and campaign news
  - Officers bought each step, sold their commissions on retirement, and could exchange into another regiment at the same rank. In 1854 an infantry major's commission cost £3,200, eleven years of his pay.

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1. Artifacts

Findspot
Found/Acquired: Benin City
Africa: sub-Saharan Africa: Nigeria: Edo State: Benin City

Materials
brass

Acquisition name
Donated by: Secretary of State for Foreign Affairs

Acquisition date
1898

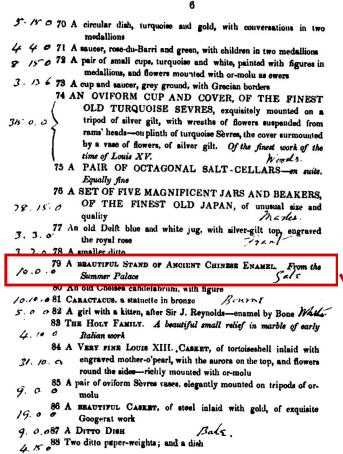
Acquisition notes
Following the British occupation of Benin City (Edo) in 1897 objects made of brass, ivory and wood were looted by British forces from the royal palace, its storerooms and compounds.



- I scrape all 1.5 million artifacts with geographical information from the BM online catalog and parse all fields including acquisition year, source, location of finding.
- Geolocate each artifact using LLM with web search on find spot.
- Aggregate to populated cities (1,941 with artifacts) and $1^\circ \times 1^\circ$ grid cells (2,073 have artifacts) by year of acquisition for 1816–1913, excluding UK and Irish places.

▷ full record ▷ what it measures ▷ coverage ▷ provenance text
▷ geocoding audit ▷ other geocodings

1. Artifacts (continued)



6

5. 10 0 70 A circular dish, turquoise and gold, with conversations in two medallions

4 4 0 71 A saucer, rose-du-Barri and green, with children in two medallions

8 15 0 72 A pair of small cups, turquoise and white, painted with figures in medallions, and flowers mounted with or-molu as ewers

3. 13 6 73 A cup and saucer, grey ground, with Grecian borders

74 AN OVIFORM CUP AND COVER, OF THE FINEST OLD TURQUOISE SEVRES, exquisitely mounted on a tripod of silver gilt, with wreaths of flowers suspended from rams' heads—on plinth of turquoise Sèvres, the cover surmounted by a vase of flowers, of silver gilt. *Of the finest work of the time of Louis XV.* Woods.

35. 0. 0

75 A PAIR OF OCTAGONAL SALT-CELLARS—*en suite. Equally fine*

76 A SET OF FIVE MAGNIFICENT JARS AND BEAKERS, OF THE FINEST OLD JAPAN, of unusual size and quality Baches.

78. 15. 0

3. 3. 0 77 An old Delft blue and white jug, with silver-gilt top, engraved the royal rose Frank

78 A smaller ditto

10. 0. 0 79 A BEAUTIFUL STAND OF ANCIENT CHINESE ENAMEL. *From the Summer Palace* Sale

80 An old Chelsea candelabrum, with figure

10. 10. 0 81 CARACTACUS, a statuette in bronze Bowne

5. 0 0 82 A girl with a kitten, after Sir J. Reynolds—enamel by Bone White

83 THE HOLY FAMILY. *A beautiful small relief in marble of early Italian work*

4. 10 0

84 A VERY FINE LOUIS XIII. CASKET, of tortoiseshell inlaid with engraved mother-o'pearl, with the aurora on the top, and flowers round the sides—richly mounted with or-molu

31. 10. 0

9. 0 0 85 A pair of oviform Sèvres vases, elegantly mounted on tripods of or-molu

86 A BEAUTIFUL CASKET, of steel inlaid with gold, of exquisite Googerat work

19. 0 0

9. 0. 087 A DITTO DISH Bale.

4. 10 0 88 Two ditto paper-weights; and a dish

- To follow the objects that passed through the market, I digitize the 27,751 auction catalogues for 1750 to 1900 scanned in Brill's *Art Sales Catalogues Online*.
- Digitize every printed lot from 900,279 pages and generate 14.5 million lots, among which 1,803 lots taken in British campaigns, in 138 sales from 19 campaigns.

▷ coverage and prices ▷ catalogue pages

10. 0. 0 79 A BEAUTIFUL STAND OF ANCIENT CHINESE ENAMEL. *From the Summer Palace* Sale

2. Battles

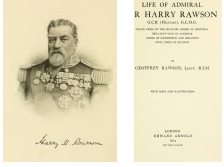
Benin ❙ 1897 ❙ British Conquest of Nigeria
Avenging the massacre at **Ugbine** (4 January), a 1,200-strong British Punitive Expedition under Admiral Harry Rawson converged on King Oba at Benin City in modern Nigeria, whose army was destroyed by Maxim guns and artillery. King Oba's capital was razed and he was exiled. His son-in-law Ologbosheri was eventually captured and executed for the attack at Ugbine (18 February 1897).

- I digitize all narratives from Jaques' *Dictionary of Battles and Sieges*. ▷ digitizing the battles ▷ coding rule and HCED
- Extract year, location, leaders, outcome, and specifically whether UK forces are involved. Manually audit every British-involved case: 474 battles, 1816–1913. ▷ other battle sources ▷ dating long battles ▷ why 1816–1913
- Similarly geolocated and aggregated to cities (219 with battles) and $1^\circ \times 1^\circ$ grid cells (231 with battles) by battle year. Main sample used all units with either artifacts or battles. ▷ maps ▷ data description ▷ summary statistics

3. Commanders

[167] **RAWSON**, SIR HARRY HOLDSWORTH (1843–1910), admiral, second son of Christopher Rawson of Woolwich, J.P. for Surrey, was born at Walton-on-the-Hill, Lancashire, on 5 Nov. 1843. He was at Marlborough College from Feb. 1854; to Christmas 1855. Entering the navy on 9 April

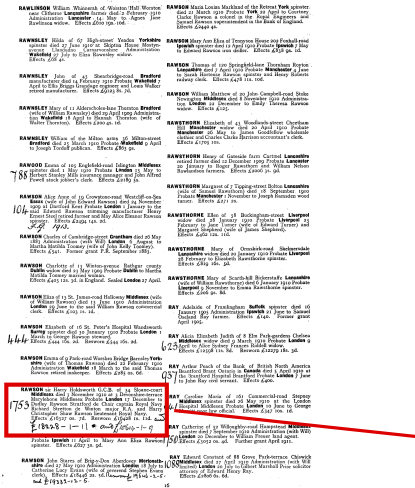
Dictionary of National Biography, 1912 supplement



Life of Admiral Sir Harry Rawson, 1914

- I identify 317 battle leaders across the 474 battles from the battle narratives and manually search their career and background information. ▷ construction
- Sources include the *Oxford Dictionary of National Biography*, *Encyclopædia Britannica*, the National Army Museum, Royal Museums Greenwich, and the Imperial War Museum, with Ancestry, FindAGrave, and WikiTree for dates of birth and death.
- I document ranks at battle and death, titles, dates of birth and death, and family background.

3. Commanders (continued)


1910.

RAWLINSON William Thomson 1st Baron of Trent Hall Worsley late Colonel **Lancaster** 1 Feb 1910 at Aldershot Administration **Lancaster** 14 May to Agnes Jane Rawlinson widow. Effects £600 19s. 10d.

RAWNSLEY Hilda of 67 High-street Yeadon **Yorkshire** spinster died 27 June 1910 at Skipton House Skipton-in-Craven Yorkshire Correspondent Administration **Wakefield** 2 July to Miss Rawnsley widow. Effects £68 4s.

RAWNSLEY John of 45 Shrewsbury-road **Bradford** manufacturer died 14 February 1910 Probate **Wakefield** 7 April to Ellis Briggs Brayshaw engineer and Louis Walker retired manufacturer. Effects £3105 8s. 7d.

RAWNSLEY Mary of 11 Aldersbrook-lane Thornton **Bradford** widow died 19 March 1910 Probate Leeds Administration (w.) **Wakefield** 18 April to Harold Hamilton (wife of Walter Thornton). Effects £200 4s. 1d.

RAWNSLEY William of the Milne acres 56 Silsbridge-lane **Bradford** died 25 March 1910 Probate **Wakefield** 9 April to Joseph Tordoff publican. Effects £803 5s.

RAWOOD Emma of 129 Englefield-road Islington **Middlesex** spinster died 1 May 1910 Probate **London** 13 May to Herbert Stanley Mills accountancy manager and John Alfred Powell stock jobber's clerk. Effects £2184 5s.

RAWSON Mary Anne of 19 Coventon-road Westcliff-on-Sea Essex widow of John Edward Rawson) died 24 November 1909 at Dartford Kent Probate **London** 1 January to the Rev. Arthur Rawson clerk in holy orders and Henry Ernest Rawson retired farmer and Mrs Alice Eleanor Rawson spinster. Effects £1294 14s. 8d.

RAWSON Charles of Cambridge-street **Dearham** died 26 May 1887 Administration (with Will) **Lancaster** 4 August to Martha Matilda Toomey (wife of John Kelly Toomey). Effects £541. Former grant P.E. September 1887.

RAWSON Charlotte of 11 Weston-grove Bathgate county **Haddington** widow died 15 May 1909 Probate **Dublin** to Martha Matilda Toomey married woman. Effects £405 10s. 3d. in England. Sealed **London** 27 April 1910.

RAWSON Eliza of 13 St. James-road Holloway **Middlesex** (wife of William Rawson) died 11 June 1910 Administration **London** 28 July to the said William Rawson commercial clerk. Effects £163 1s. 1d.

RAWSON Elizabeth of 16 St. Peter's Hospital Wandsworth **Surrey** spinster died 30 January 1910 Probate **London** 1 March to George Barrett surveyor. Effects £444 6s. 2d. Resworn £444 6s. 2d.

RAWSON Emma of 3 Park-road Weedon Bridge Recruits **York** widow died 15 February 1910 at Harrogate Letters of Administration **Wakefield** 18 March to Ida and Thomas Rawson retired innkeeper. Effects £985 9s. 6d.

RAWSON sir Harry Holdsworth G.C.B. of 34 Sloane-court **Middlesex** died 3 November 1910 at 3 Devonshire-terrace Marylebone Middlesex Probate **London** 17 December to Dudley Rawson Stratford de Chair captain Royal Navy Richard Stretton de Winton major R.A. and Harry Christopher Shaw Rawson lieutenant Royal Navy. Effects £10327 0s. 7d. Resworn £10428 1s. 11d.

RAWSON Mary Ann Eliza of Tempest House 202 Penhall-road **Lanark** spinster died 12 April 1909 Probate **Lanark** 7 May to Edward Rawson son duties. Effects £838 9s. 1d.

RAWSON Thomas of 190 Springfield-lane Thornton Royton **Lancaster** died 2 December 1909 Administration **London** 19 March Harrison Rawson and Henry Elsdon railway clerk. Effects £478 11s. 10d.

RAWSON William Matthew of 20 John Campbell-road Stoke Newington **Middlesex** died 8 December 1910 Administration **London** 18 December to Emily Theresa Rawson widow. Effects £102.

RAWSTHORN Elizabeth of 45 Woodlands-street Clayton-le-Moors **Lancaster** widow died 27 May 1910 Probate **Blackburn** 28 May to James Cockshott solicitor's clerk and Charles Clarke Harrison accountant's clerk. Effects £1769 10s.

RAWSTHORN Henry of Cuerdale farm Cuerdale **Lancaster** retired farmer died 21 December 1909 Probate **Lancaster** 7 February to Robert Rawsthorn and John Peter Rawsthorn farmers. Effects £2006 3s. 3d.

RAWSTHORN Margaret of 7 Tipping-street Bolton **Lancashire** (wife of Samuel Rawsthorn) died 18 September 1910 Probate **Liverpool** 7 November to Joseph Rawsthorn wood turner. Effects £402 1s.

RAWSTHORNE Ellen of 38 Buckingham-street **Liverpool** widow died 28 January 1910 Probate **Liverpool** 23 February to Ellen Rawsthorne (wife of Edward Turner) and Margaret Rawsthorne (wife of James Shepherd). Effects £460 12s. 11d.

RAWSTHORNE Mary of Ormskirk-road Skelmersdale **Lancashire** widow died 20 January 1910 Probate **Liverpool** 28 February to Elizabeth Rawsthorne spinster. Effects £62 16s. 3d.

RAWSTHORNE Mary of Scarth-hill Rainford-le-Rainford **Lancashire** (wife of William Rawsthorne) died 7 January 1910 Probate **Liverpool** 9 November to Emma Rawsthorne spinster. Effects £206 3s. 8d.

RAY Adelaide of Framlingham **Suffolk** spinster died 16 December 1909 at Framlingham aforesaid Probate **Ipswich** 11 June to Samuel Ouchard Ray farmer. Effects £146. Former grant April 1901.

RAY Alice Elizabeth Judith of 8 Elm Park-gardens Chelsea **Middlesex** widow died 9 March 1910 Probate **London** 3 August to Alice Isobel Frances Riddell widow. Effects £14598 11s. 8d. Resworn £14259 18s. 3d.

RAY Arthur Punch of the Bank of British North America Lombard-street Ockham in **Surrey** died 1 April 1910 at the Stratford Hospital Stratford Probate **London** 7 June to John Ray coal merchant. Effects £4260.

RAY Caroline Maria of 161 Commercial-road Stepney **Middlesex** spinster died 26 May 1910 at the London Hospital Whitechapel Probate **London** 15 June to George Ray coal merchant's clerk. Effects £241 10s. 1d.

RAY Catherine of 12 Willoughby-road Hampstead **Middlesex** spinster died 7 September 1910 Administration (with Will) **London** 11 October to William Fraser Reid agent. Effects £3052 0s. 2d. Further grant April 1911.

RAY Edward Courtnay of 88 Grove Park-terrace Chiswick **Middlesex** died 12 July 1910 Probate **London** 29 August to Edward Landon de Ville solicitor Alexander Ray solicitor attorney of Edward Mary Ray. Effects £6060 6s. 6d.

RAWSON sir Harry Holdsworth G.C.B. of 34 Sloane-court **Middlesex** died 3 November 1910 at 3 Devonshire-terrace Marylebone Middlesex Probate **London** 17 December to Dudley Rawson Stratford de Chair captain Royal Navy Richard Stretton de Winton major R.A. and Harry Christopher Shaw Rawson lieutenant Royal Navy. Effects £10327 0s. 7d. Resworn £10428 1s. 11d. and £18228 – 1 – 11 and £8614 – 1 – 9

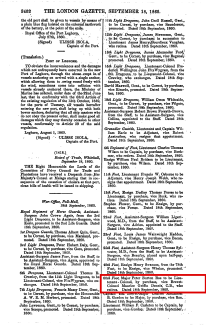
- I digitize the England and Wales probate calendars, 1858–1996 (16 million records), supplemented by Scottish and Irish records.
- I extract name, death year, address, occupation (ends 1900s), and estate value.
- I manually search and link battle leaders to probates: of the 245 within calendar coverage, 218 (89%) are matched.

▷ digitization and linkage ▷ bounds for unlinked commanders

4. Prize and Rank Purchasing

ROYAL WARRANT GRANTING PRIZE MONEY TO THE "SAUGOR FIELD FORCE" UNDER THE COMMAND OF MAJOR-GENERAL WHITLOCK.

Booty, or have been otherwise realized, or are about to be realized, amounting, or computed to amount, with interest, after deducting costs, to the sum of fifty-five lacs, fifty thousand rupees, or thereabouts: And whereas the said Commis-



- I digitize all London Gazettes until 1914 (28k issues and 656k pages) and extract all officer movement information for 1816–1870 (when commission purchase was permitted).
- Extracted 169,449 Army movement records including promotions, exchanges, and retirements.
- The Gazette also records prize-money distributions. For the Army, I identify 24 prize funds with Gazette notices for wars fought between 1816 and 1913.

▷ record ▷ extraction audit ▷ prize funds

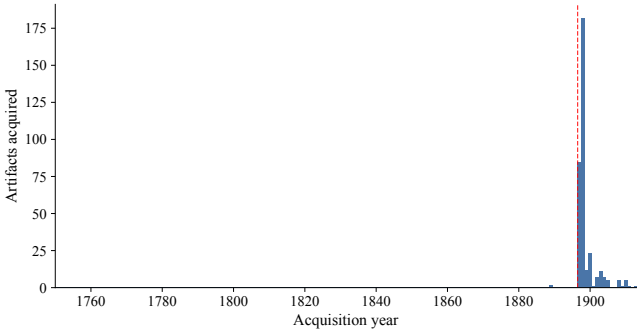
60*th Foot*, Major Peter Burton Roe to be Lieutenant-Colonel, by purchase, vice Brevet-Colonel Maurice Griffin Dennis, C.B., who retires. Dated 18th September, 1860.

Roadmap

1. Background
2. Data
3. **Artifacts**
4. Officers
5. Conclusion

Motivating Case: Benin City (Nigeria), 1897

Almost nothing before 1897, a spike right after the punitive expedition, then a long tail as private holdings reached the market.



▷ other cases ▷ channels and timing ▷ Benin to 2024

Empirical Strategy

- Does a British battle in city i in year $t-1$ raise acquisitions from i in year t? I estimate the following Poisson pseudo-maximum likelihood (PPML) model:

$$E[y_{it} \mid BritishBattle_{i,t-1}] = \exp\left(\beta\, BritishBattle_{i,t-1} + \alpha_i + \lambda_t + W'_{it}\gamma\right)$$

- y_{it}: artifacts from city i acquired in year t, a sparse count.
- α_i: city effects (a stock of antiquities). λ_t: year effects (the Museum's budget).
- W_{it}: country $\times$ year and size-decile $\times$ year effects.

Identifying assumption

Conditional on the fixed effects, the timing of a battle within a place was unrelated to the acquisitions the Museum would otherwise have made from it.

▷ tests of this concern ▷ why PPML ▷ other estimators

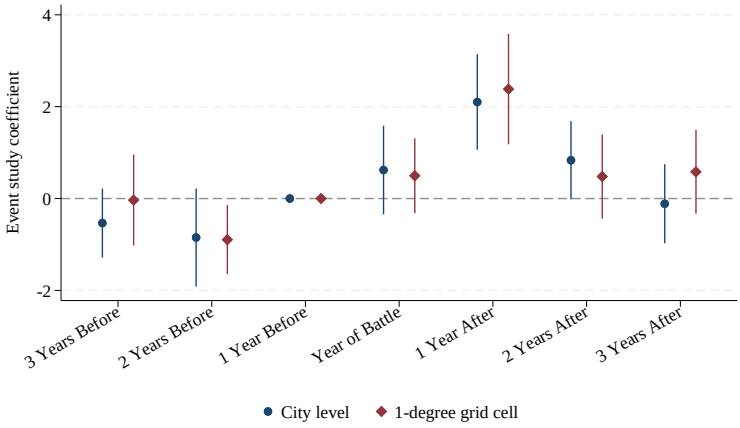
Main Results

A battle multiplies next-year acquisitions from its city by $e^{2.15} \approx 8.6$ (column 1) to $e^{2.83} \approx 17$ (column 3). The baseline suggests an 8.6-fold rise, from 0.7 artifacts to about 6.

	Number of Artifacts Acquired This Year					
	(1)	(2)	(3)	(4)	(5)	(6)
Previous Year Had Any British Battle	2.150***	2.128***	2.833***	2.413***	2.464***	3.260***
	(0.534)	(0.378)	(0.342)	(0.605)	(0.777)	(0.574)
Unit of observation	City	City	City	1° Grid	1° Grid	1° Grid
City or Grid FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	–	–	Yes	–	–
Country × Year FE	No	Yes	Yes	–	–	–
Pop./Area Decile × Year FE	No	No	Yes	–	–	–
10° Block × Year FE	–	–	–	No	Yes	Yes
Cell Pop./Settlement Decile × Year FE	–	–	–	No	No	Yes
Number of Observations	198,450	198,450	198,450	213,248	213,248	213,248
Mean of dep. var.	0.704	0.704	0.704	0.655	0.655	0.655
SD of dep. var.	18.460	18.460	18.460	17.778	17.778	17.778

Event Study

$$E[y_{it} \mid B_i] = \exp\left(\sum_{k=-3, k\neq -1}^{3} \beta_k \mathbf{1}[t-k \in B_i] + \alpha_i + \lambda_t\right)$$



Threats to Identification

Concern: a local shock, such as growing trade or a revolt, that made a battle more likely and moved objects anyway.

Accounting for common trends

- A shock already at work before the battle year? Flat pre-trends, robust to unit trends, and recent acquisitions do not predict a British battle. ▷ unit trends ▷ prediction test

Battles triggered by anti-British incidents

- A shock arriving with the battle? Acquisitions still rise after 34 battles set off by anti-British incidents. ▷ the 34 battles ▷ why not IV

Falsification

- Warfare in general, or chance? Falsification tests. ▷ did Britain fight for the spoils?

Battles Triggered by Anti-British Incidents

- 34 battles were set off by a prior anti-British incident: an envoy killed, a Residency massacred, a treaty repudiated, or British shipping seized.

▷ the 34 battles

▷ why not an instrument

	Number of Artifacts Acquired This Year					
	(1)	(2)	(3)	(4)	(5)	(6)
Previous Year Had Any British Battle	2.150***			2.413***		
	(0.534)			(0.605)		
Previous-Year Battle, Incident-Triggered		2.397**	3.335***		3.685***	5.575***
		(0.950)	(0.673)		(0.999)	(1.027)
Previous-Year Battle, Other		2.075***	2.680***		1.772***	2.491***
		(0.623)	(0.399)		(0.616)	(0.652)
Unit of observation	City	City	City	1° Grid	1° Grid	1° Grid
Estimation	PPML	PPML	PPML	PPML	PPML	PPML
City or Grid FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	Yes	Yes	No
Country × Year FE	No	No	Yes			
Pop./Area Decile × Year FE	No	No	Yes			
10° Block × Year FE				No	No	Yes
Cell Pop./Settlement Decile × Year FE				No	No	Yes
Number of Observations	198,450	198,450	198,450	213,248	213,248	213,248
Mean of dep. var.	0.704	0.704	0.704	0.655	0.655	0.655
SD of dep. var.	18.460	18.460	18.460	17.778	17.778	17.778

Falsification: Other Battles and Defeats

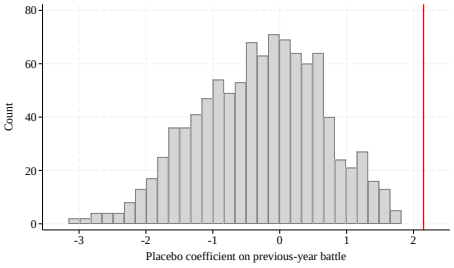
	Number of Artifacts Acquired This Year					
	(1)	(2)	(3)	(4)	(5)	(6)
Previous Year Had Any British Battle	2.150***			2.413***		
	(0.534)			(0.605)		
Previous Year Had Non-British Battle		-0.864*			-0.532	
		(0.467)			(0.389)	
Previous Year Had British Battle Won			2.180***			2.454***
			(0.544)			(0.609)
Previous Year Had British Battle Lost			-1.877*			-1.322
			(1.097)			(0.868)
p-value: won = lost			0.001			0.000
Unit of observation	City	City	City	1° Grid	1° Grid	1° Grid
City or Grid FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Number of Observations	198,450	198,450	198,450	213,248	213,248	213,248
Mean of dep. var.	0.704	0.704	0.704	0.655	0.655	0.655
SD of dep. var.	18.460	18.460	18.460	17.778	17.778	17.778

▷ civil-war and French-only battles

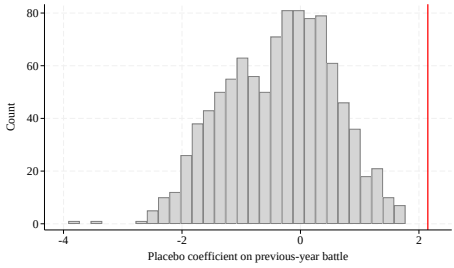
Falsification: Randomly Reassigned Battles

Reassigning each battle's year among the actual battle years, or its city among the actual battle cities gives medians of -0.24 and -0.27. The baseline 2.15 exceeds every draw.

Battle years shuffled



Battle-city hosts shuffled



Robustness

Comparisons under recurring treatment

- PPML with stacked difference-in-differences: each battle compared only with cities without a battle in nearby years. ▷ design ▷ comparison groups
- Local-projections difference-in-differences: clean comparisons at each horizon. ▷ estimates ▷ event studies

Functional form

- OLS in levels, logs, and inverse hyperbolic sines, and a negative binomial. ▷ functional form
- Any acquisition. ▷ extensive margin

Outliers

- Winsorize, trim, or top-code, and drop top cities or any country. ▷ outliers

Alternative treatments

- Regular forces only, HCED coding, Wikidata battles. ▷ battle sources
- Start, end, or random year of long battles. ▷ dates

Alternative outcomes

- Imprecise locations spread, precise only, production place. ▷ geocoding
- Ever exhibited, no private donations. ▷ outcomes

Sample, spatial spillovers, and inference

- Cities existing by 1815 or 1900, back to 1753. ▷ samples
- State, 2°, and 5° units, and Conley standard errors. ▷ Conley

Long-Run Dynamics: Design

I use a staggered difference-in-differences design to follow acquisitions after a city's first British battle.

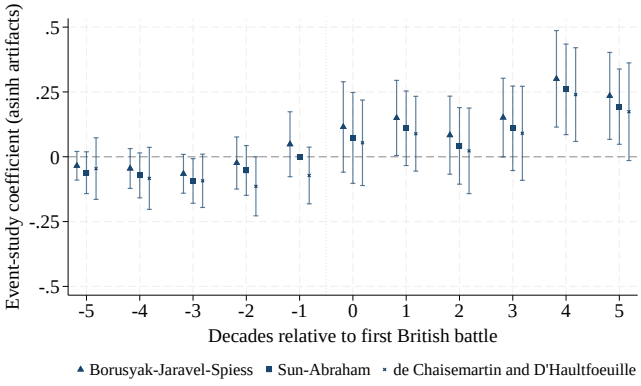
$$\tilde{y}_{id} = \sum_{h=-5}^{+5} \theta_h \mathbf{1}[d - g_i = h] + \alpha_i + \lambda_d + \varepsilon_{id}$$

- A city-decade panel, 1750–2024: $\tilde{y}_{id} = \operatorname{asinh}(y_{id})$, where y_{id} counts acquisitions from city i in decade d.
- g_i is the decade of the first British battle during 1816–1913. Exposure continues thereafter, and later battles do not reset the clock.
- City and decade effects. I compare with cities without a British battle and, where permitted by the estimator, cities before their first battle.

I report Borusyak–Jaravel–Spiess, Sun–Abraham, and de Chaisemartin–D'Haultfœuille estimates.

Long-Run Dynamics

The Museum kept receiving artifacts from battle cities for decades afterward.



▷ long-run comparison ▷ raw decade averages ▷ pooled estimates ▷ synthetic DiD ▷ any acquisition ▷ arrival lags

Mechanisms: Military Access

- The rise is 1.24 where Britain already ruled and 3.88 where it never ruled.
- It is also larger in countries with weaker military institutions and a richer stock of material culture.

Dependent variable:	Number of Artifacts Acquired This Year		
	(1)	(2)	(3)
Sample:	Already Colony	Later Colony	Never Colony
Previous Year Had Any British Battle	1.236**	0.651	3.883***
	(0.501)	(0.920)	(0.600)
Unit of observation	City	City	City
City FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Number of Observations	39,533	30,598	127,596
Mean of dep. var.	1.300	0.394	0.597
SD of dep. var.	30.019	7.167	15.439

▷ measures ▷ institutions and material culture ▷ the two country measures ▷ British-rule indicator

The Museum's Own Records

Military source:
Source names British military personnel



Acquisition name

Purchased from: Capt C Johnson

War-related:
Notes link the object to a war



Acquisition notes

Af1909,0315.1 to 4 are all listed in the acquisition register as coming 'from the fields of Omdurman and the Atbara'. Lady Gatacre's late husband had been involved in the 1898 war.

Explicit loot:
Notes explicitly describe wartime taking



Acquisition notes

Following the British occupation of Benin City (Edo) in 1897 objects made of brass, ivory and wood were looted by British forces from the royal palace, its storerooms and compounds.

Mechanisms: Wartime Taking

The effect grows as the records name a military source, a war, or a taking.

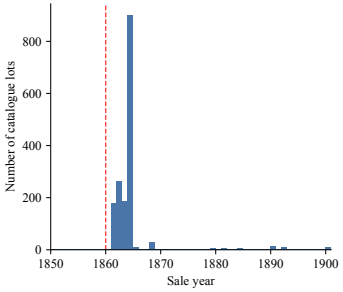
Dependent variable:	Number of Artifacts Acquired This Year				
Sample:	(1) Full sample	(2) Military source	(3) War-related	(4) Explicit loot	(5) None of (2)–(4)
Previous Year Had Any British Battle	2.150***	2.654***	4.237***	4.693***	2.066***
	(0.534)	(0.973)	(0.462)	(1.132)	(0.638)
Unit of observation	City	City	City	City	City
City FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Number of Observations	198,450	198,450	198,450	198,450	198,450
Mean of dep. var.	0.704	0.036	0.010	0.002	0.661
SD of dep. var.	18.460	3.139	0.761	0.455	18.154

▷ what the labels cover ▷ example notes ▷ city and grid ▷ how the sets are built

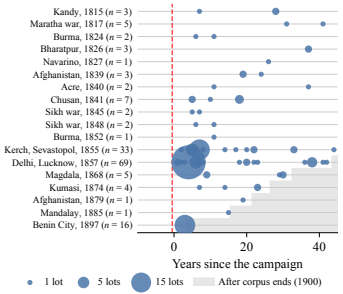
Mechanisms: Turning Spoils into Cash

- I identify 33 prize funds for named campaigns, and the 20 with a stated total add up to about £3.4 million, or US$700 million at 2025 prices.
- 1,803 lots sold as taken in war, from 19 campaigns, over decades.

Yuanmingyuan lots by sale year



Other campaigns, years since the campaign



▷ prize sales ▷ prize funds ▷ salerooms ▷ catalogue pages ▷ salerooms and Museum ▷ coverage and prices

The Scale of Battle-Related Acquisitions

- About 1,570 artifacts in the battle year and the three after it. ▷ calculation inputs
  - 18.2% of acquisitions from battle cities in 1816–1913.
  - 16.7% of them exhibited, against 5.1% of the rest.
- Using a multilingual BERT model fine-tuned on 261,122 priced auction lots from the Getty Provenance Index and *Art Sales Catalogues Online*, the artifacts attributed to battles in the battle year and the three years after it sum up to £15,600 at 1900 prices (US$3.6 million today), 21% of the valuation of the Museum's acquisitions from battle cities. ▷ valuation model
- An understatement: four years only, saleroom rates of the time, and the Museum one destination among salerooms and estates. ▷ arrival lags

Roadmap

1. Background
2. Data
3. Artifacts
4. **Officers**
5. Conclusion

Did the Spoils Reach the Officers?

- Did commanders of battles that yielded spoils die richer? After a war rich in spoils, did more officers pay to join and advance in its regiments?
- Sort battles into two types — **artifact-heavy** (BM inflow from the battle's city jumps after the battle) versus **artifact-scarce** (little or no change).
- Various definitions: strict (post period is $2\times$ pre, or if pre-period is zero, post period > 5), any post-inflow, or above-median-given-inflow, at 3-year (immediate response) and 20-year (slow release) windows.

▷ classification examples

How Were Commanders Assigned?

- Commanders required appointment by the government or military superiors.(Biddulph, 1904).
- Existing postings often determined who was available. In 31% of the commander–battle pairs, commanders already held a standing command in or next to the theater.
- Rawson had commanded the Cape of Good Hope and West Africa Station since 1895, two years before the events that triggered the Benin expedition.
- Importantly, the importance of battles do not suggest the spoil prospect of the battles.

▷ assignment evidence and cases

Commanders' Wealth: Design

- I link the 317 commanders (474 battles) to probate records by name, year of death, and rank: 245 died within calendar coverage ($\geq$1858); 218 (89%) matched.
- X_i fixed before the battles: rank at battle, age, birthplace, noble family, military father, regiment type. Rank at death, peerage, and honors can follow a profitable campaign, so they do not enter X_i.
- Identify an OLS model with controls:

$$\text{Wealth at death}_i = \beta \cdot (\#\,\text{Artifact-Heavy Battles})_i + X_i'\gamma + \epsilon_i$$

▷ threats: ability and choice ▷ assignment evidence and cases

Probated Wealth

Dep. Var.: Probate Wealth at Death (£ thousand, 1900 prices)	(1)	(2)
Panel A: 3-Year Window		
Number of artifact-heavy (strict 2× pre) battles	12.252*	14.380*
	(6.848)	(7.704)
Number of artifact-heavy (any post-inflow) battles	7.501***	9.238***
	(2.526)	(3.084)
Number of artifact-heavy (above median, inflow > 0) battles	13.594**	15.381**
	(6.837)	(7.612)
Panel B: 20-Year Window		
Number of artifact-heavy (strict 2× pre) battles	12.103***	13.390***
	(4.155)	(4.811)
Number of artifact-heavy (any post-inflow) battles	6.863***	7.432***
	(2.044)	(2.187)
Number of artifact-heavy (above median, inflow > 0) battles	10.904***	12.068***
	(3.525)	(4.052)
Background and Rank Controls	No	Yes
Number of Observations	218	218
Mean of dep. var. (£k, 1900)	20.42	20.42
SD of dep. var. (£k, 1900)	35.37	35.37

Who Are the Other Officers?


STANHOPE Philip Spencer.
313 Personal Estate under £60,000.
Resworn June 1880 under £50,000.

12 March. The Will with a Codicil of Philip Spencer Stanhope late of 70 Harley - street in the County of Middlesex a General in Her Majesty's Army and Colonel of the 13th Prince Albert's Light Infantry Regiment who died

- The 1858–1906 probate calendars record occupation as well as estate value. I identify military officers from ranks in their occupation descriptions.
- I exclude records matching a battle commander by name and year of death.
- This leaves 13,556 other officers with usable wealth and rank information. I compare them with 140 commanders, controlling for rank at death and death decade.

Commanders and Other Officers

I compare 140 commanders with 13,556 other officers recorded in the 1858–1906 probate calendars. Define Artifact-heavy commanders as who led at least one strict artifact-heavy battle, using the 20-year window.

Dep. Var.:	Probate Wealth at Death (£ thousand, 1900 prices)		
	(1)	(2)	(3)
Any battle commander vs. Officers in probate	17.238***	14.773***	15.233***
	(3.413)	(3.372)	(3.391)
Artifact-heavy battle commanders vs. Officers in probate	33.693***	31.767***	31.871***
	(12.332)	(12.180)	(12.218)
Other battle commanders vs. Officers in probate	12.939***	10.318***	10.859***
	(2.706)	(2.689)	(2.702)
Rank-Group-at-Death FE	No	Yes	Yes
Death-Decade FE	No	No	Yes
Number of Observations	13,696	13,696	13,696
Mean of dep. var. (£k, 1900)	7.93	7.93	7.93
SD of dep. var. (£k, 1900)	26.17	26.17	26.17

Probated Wealth

- 3-year window: each additional artifact-heavy battle is associated with roughly £7,500–£15,000 more probated wealth.
- 20-year window, which allows for delayed release: roughly £7,000–£13,000, 34 to 66 percent of the £20,400 mean.
- Compared with other officers in probate, commanders of artifact-heavy battles left about £31,900 more, versus £10,900 for other commanders, controlling for rank at death and death decade.

Probated Wealth: Robustness

- Winsorized or trimmed estates. ▷ winsorize, trim
- Logs and Poisson. ▷ logs, Poisson
- Accounting windows of 2 to 5 and 20 to 50 years. ▷ 2–50 year windows
- Extensive and intensive margins. ▷ extensive and intensive margins
- Lee bounds on the probate linkage. ▷ Lee bounds
- Dropping one commander at a time. ▷ drop one commander
- Commanders against other officers in probate. ▷ other officers

How Commanders Got Rich: Prize Money

- I obtain a prize-money figure for 29 of the 218 commanders. Where the amount is documented, it equals 40% of wealth at death on average.
- Colin Campbell, son of a Glasgow cabinetmaker, was entitled to about £30,000 from four Mutiny funds, and died with £102,000, the seventh largest of the 218 estates.
- An understatement: booty settled on the spot never reached the *Gazette*.

▷ all 29 prize receipts ▷ sources and gaps

How Commanders Got Rich: Careers

- On the 20-year window, an additional artifact-heavy battle raised the chance of a peerage at death by 12.9 percentage points (mean 7.3%).
- It raised the chance of high office, a governorship or a command-in-chief, by 9.3 percentage points (mean 28%). ▷ careers table
- Time to realize gains: commanders who died within three years of their last battle left smaller estates. ▷ death timing
- Other returns: objects officers kept and sold.

Commission Purchases after the Indian Mutiny

- The Mutiny (1857–58) paid about £1.2 million in prize under eleven Royal Warrants. The Banda and Kirwee booty alone was valued at £700,000, and its first distribution gave a private £50, nearly three years of his pay.
- Two comparisons:
  - Strict: 25 regiments whose main campaign was the Mutiny, against 31 with no substantial combat.
  - Broad: all 38 regiments with Mutiny service, against all other purchasable regiments.

$$y_{rt} = \sum_{k \neq 1858} \beta_k \, \mathit{Treated}_r \cdot \mathbf{1}[t = k] + \alpha_r + \lambda_t + \varepsilon_{rt}$$

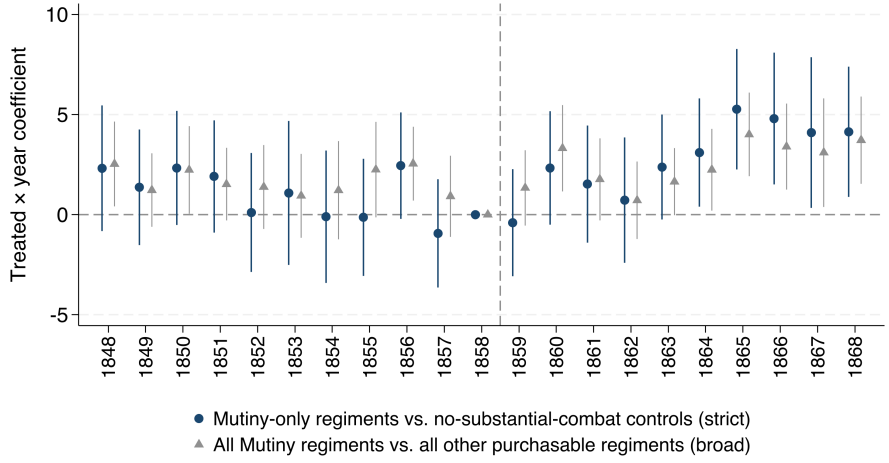
- y_{rt}: purchase events in regiment r, year t, 1848–1868. Clustered by regiment.

▷ why the Mutiny ▷ the regiments ▷ battle honors ▷ a contemporary account ▷ Afghan War

Regiments and Repeated Campaigns

- Regiments were continuing army organizations. Officers held commissions in particular regiments, whose battalions could serve in successive campaigns (Bruce, 1980).
- Regiments rotated between home and overseas service, and the usual expectation is an regiment stay overseas for 10 years. But military needs and other arrangements could delay their return, and overseas tours could exceed two decades (Hansard, 1860, 1866).
- Some regiments served in the Mutiny operations went on to fight in the Second Opium War in 1860 (Hutton, ed, 1912).

Commission Purchases



▷ DiD table ▷ robustness ▷ comparisons ▷ specifications ▷ drop one regiment ▷ Afghan War ▷ Afghan design

Who Bought? Attraction, Cash, Release

	Gazetted Officer-Career Events per Regiment-Year					
	Baseline	Attraction		Cash in hand	Release	
	(1) All purchase events	(2) Exchanges into the regiment	(3) First commissions purchased	(4) Within-regiment promotions by purchase	(5) Retirements and resignations	(6) Purchases into retirement vacancies
Panel A: Mutiny-only regiments vs. no-substantial-combat controls (strict)						
Treated × Post (1859–68)	1.850**	0.457***	0.797**	1.104**	0.613*	0.573*
	(0.829)	(0.166)	(0.379)	(0.476)	(0.364)	(0.306)
Number of Observations	1,176	1,176	1,176	1,176	1,176	1,176
Mean of dep. var.	5.91	1.31	2.36	3.47	2.60	2.23
Panel B: All Mutiny regiments vs. all other purchasable regiments (broad)						
Treated × Post (1859–68)	1.001*	0.324**	0.367	0.650*	0.371	0.337
	(0.589)	(0.134)	(0.263)	(0.348)	(0.259)	(0.214)
Number of Observations	2,667	2,667	2,667	2,667	2,667	2,667
Mean of dep. var.	5.96	1.17	2.39	3.49	2.63	2.24
Regiment FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Who Bought? Attraction, Cash, Release

- Outsiders: exchanges in +45%, first commissions +50%. They held no prize from the campaign, so their entry reflects attraction.
- Incumbents: 1.10 more purchases within the regiment per year, the largest part, which could reflect prize money or the same attraction.
- Retirement vacancies: 0.57 of the 1.85, less than a third.

▷ both comparisons ▷ event studies by channel

Spoils drew officers toward the regiments that had earned them, and the officers who had earned them bought their next steps.

Roadmap

1. Background
2. Data
3. Artifacts
4. Officers
5. **Conclusion**

Conclusion: Spoils as Military Compensation

1. Battles systematically yielded spoils. Museum acquisitions rose after British battles and remained higher for decades.
2. Officers benefited financially. Commanders of spoil-rich battles left larger estates, and prize payments could amount to years of salary.
3. Officers paid to enter and advance. After the Indian Mutiny, commission purchases rose by 45% in participating regiments.

Official salaries understated the financial rewards of imperial service.

Broader Implications

- Compensation can include rights to the proceeds of organizational activity. The state rewarded military service with captured wealth, shifting part of its cost onto local societies.
- Beyond Britain, the Louvre, Rijksmuseum, and Hermitage offer scope to study whether other empires also rewarded military service with spoils.
- Future research can compare salary payments with rights to spoils, asking whether alternative arrangements could reward service with lower social costs under fiscal constraints.

Related Literature

- Conflict: states and economies (Besley and Persson, 2009; Dincecco and Prado, 2012; Voigtländer and Voth, 2013). Individuals choose between producing and taking (Dal Bó and Dal Bó, 2011). Potential gains of armed groups (Collier and Hoeffler, 2004; Dube and Vargas, 2013; Berman et al., 2017).
- Two readings of British military institutions: officers as salaried servants and looting as indiscipline (Smith, 1776; Tilly, 1990; Sandholtz, 2007), or purchase and prize as an incentive contract (Allen, 1998).
- Returns of empire: to taxpayers (Davis and Huttenback, 1986; O'Brien, 1988), to investors (Offer, 1993; Bogart and Chaudhary, 2019), to the economy (Heblich et al., 2022; Dal Bó et al., 2025). Returns to office (Eggers and Hainmueller, 2009; Querubín and Snyder, 2013; Guardado, 2018; Xu, 2018).
- What empire cost the colonized (Acemoglu et al., 2001; Nunn, 2008; Dell, 2010; Huillery, 2014).
- Art and antiquities: Ashenfelter and Graddy (2003), Oosterlinck (2017), Fisman and Wei (2009), Kremer and Wilkening (2025). Museums and collecting: MacKenzie (2009), Longair and McAleer (2012), Hicks (2020), Patrizio Gunning and Challis (2023).
- State capacity: motivating agents (Besley and Persson, 2009; Finan et al., 2017) and non-salary compensation (Johnson and Koyama, 2014; Hillmann and Gathmann, 2011). ▷ go

▷ Back

State Capacity and Non-Salary Compensation

- How states motivate the agents who act for them (Besley and Persson, 2009, 2011; Finan et al., 2017; Besley et al., 2022). Pay, promotion, postings, and recognition select and motivate agents (Dal Bó et al., 2013; Bertrand et al., 2020; Khan et al., 2019; Ager et al., 2022), and patronage shapes how they perform (Xu, 2018; Voth and Xu, 2022).
- States had to tax and borrow to finance wars (Tilly, 1990; Brewer, 1989), and their capacity to do either was limited (Besley and Persson, 2009; Johnson and Koyama, 2014).
- Historically, states paid many agents in non-salary forms, from tax farmers to privateers (Johnson and Koyama, 2014; Hillmann and Gathmann, 2011), and later replaced them with salaried officials and state-funded forces (Tilly, 1990; Parrillo, 2013), a shift that classic accounts link to the end of plunder (Smith, 1776; Redlich, 1956).
- This paper: the largest empire of the nineteenth century still paid its officers beyond their salaries with a share of what they captured, and so met part of their compensation without drawing on state finances.

▷ Back

Can the Museum Return Objects?

- The British Museum Act 1963 prohibits the Trustees from disposing of objects except in narrow cases: duplicates, human remains, and objects looted in the Nazi era (Savoy, 2022).
- The Museum has never permanently returned a contested object in modern times. It lends, though it declined to lend the Queen Idia ivory mask to Nigeria for the 1977 Lagos festival.
- France passed Law No. 2026-351 in May 2026, which lets public bodies return objects illicitly appropriated between 1815 and 1972 (République française, 2026).
- The paper supplies magnitudes to a debate that has rested on individual cases. Two findings matter for provenance review: an institution's own labels understate the taking, and arrivals stretch over decades.

▷ Back

Officers' Pay in Context

- The army scale of 1806 ran from £96 for an ensign to £365 for a lieutenant-colonel and stayed there until 1914 (Spiers, 1980; Wood, 2011). By the late 1860s a cavalry subaltern needed a private income of at least £400 (Bruce, 1980).
- The authorities never treated pay as full remuneration. Windham told the Commons in 1806 that "none could enter the army as officers, who had not something else besides their pay" (Bruce, 1980). Fortescue called pay "rather the tools of a trade than the profits of a trade" (Harries-Jenkins, 1977).
- Benchmarks: a commercial clerk earned £201 in 1815 and £236 in 1851, an agricultural laborer £29 in 1851 (Williamson, 1982), and a curate £90–120 in the 1890s (Beckett, 2018).
- A captain's £190 is about US$32,000 at 2025 prices at the price level of 1806 and US$49,000 at that of 1851. A naval lieutenant's £110 is about US$18,000.
- Navy, 1814 scale: a lieutenant on full pay £110, an admiral commanding in chief £1,820. Of 4,000 lieutenants after the peace, 3,000 sat on the minimum half-pay of 5s a day (Lewis, 1960).

▷ Back

The Purchase Tariff and the Premiums

- Regulation tariff of 1854, infantry: ensign £450, lieutenant £700, captain £1,800, major £3,200, lieutenant-colonel £4,500. A lieutenant-colonelcy cost £6,175 in the dragoons, £7,250 in the Life Guards, and £9,000 in the Foot Guards (Bruce, 1980).
- Over-regulation premiums came on top: in the 1860s about £4,570 for the step to lieutenant-colonel in the Household Cavalry and £2,500 in the infantry, and £3,700 for a Foot Guards captaincy.
- About a third of commissions changed hands without purchase, through death vacancies and cadetships, and above lieutenant-colonel promotion went by seniority.
- Allen (1998) reads army purchase and naval prize as two solutions to incentive problems: fixed targets on land, moving targets and monitored prize at sea.
- The Cardwell reforms abolished purchase in 1871 and compensated officers for the value of their commissions.

▷ Back

How Commission Purchase Worked

- Purchase covered infantry and cavalry commissions up to lieutenant-colonel. The navy, artillery, engineers, and East India Company units had no purchase.
- Entrants began at the lowest commissioned rank. They needed a recommendation and, from 1849, a qualifying examination (Bruce, 1980).
- A retiring officer sold his commission. The promotion chain compensated him as junior officers paid to move up. Officers could also exchange into another regiment.
- About a third of commissions changed hands without purchase, mainly through death vacancies and cadetships. Above lieutenant-colonel, promotion followed seniority.

The purchase records measure paid entry and advancement within particular regiments.

▷ prices ▷ credit and campaign news ▷ Back

Purchase Finance and Campaign News

- **Finance.** Officers tied up capital in commissions. Illegal premiums, army-agent credit, and social expenses affected the cost beyond the tariff (Bruce, 1980).
- **Public reports.** The *Gazette* published prize distributions. Engels's 1858 dispatch quoted Russell of *The Times* on officers making fortunes at Lucknow (Engels, 1858).
- **Vacancies.** Officers could rarely sell on active service. After the Mutiny windfall, so many applied to retire that the Duke of Cambridge withheld approval (Bruce, 1980; Engels, 1858).
- The data record transactions, without their prices or the buyers' prize receipts. They cannot apportion incumbents' purchases between new cash and expected returns.

▷ Back

How Army Prize Worked

- No army regulation described booty. Wolseley's *Soldier's Pocket-Book* supplied the procedure because "no orders on the subject are to be found in our regulations" (Wolseley, 1886).
- Booty belongs to the Crown. The officers elect prize agents, every unit sends prize rolls to the War Office, and the booty is sold by auction on the spot or at home.
- The Treasurer of Chelsea Hospital distributes the proceeds by a scale that follows daily pay: one share for a private, 400 for a general, and 5% of the fund for the general commanding.
- Royal Warrants covered only some campaigns: Bharatpur, Ghuznee, Sindh, Mooltan, and the Mutiny funds. Elsewhere the booty was sold in the field, kept by officers, or passed to government custody.
- Sums could be large: Charles Napier's commander's eighth of the Sindh prize was about £68,000. One argument for regulated prize was discipline, since soldiers who expected a share were less likely to plunder on their own (Parliamentary Papers, 1870).

▷ Back

How Naval Prize Worked

- Since 1708, an Act of Parliament had granted captured property to the officers and men who took it, and each war's Prize Act fixed how the proceeds were split by rank (Lewis, 1960; Aldous, 2020).
- Naval prize ran through the Admiralty Court, which condemned ships and cargoes, distributed shares according to prize rules, and published notices in the *London Gazette*. The regime was statutory, reviewable, and relatively predictable (Aldous, 2020).
- The Admiralty Court Act 1840 extended the prize court's jurisdiction to booty taken on land, and the Naval Prize Act 1864 consolidated the law of prize. The Prize Act 1948 abolished prize money.
- Prize "was one of the great inducements in the naval service, especially to officers," and "even lieutenants and warrant officers could become quite wealthy, given luck" (Lavery, 1989).
- For 1816–1914 the *Gazette* carries 7,627 naval notices of distributions, head money, and bounties, covering 4,875 captures. They state the sum paid to a share class, not the value of the fund.

▷ Back

The Word “Loot”

- The English word is itself a product of plunder. British troops in India adopted the Hindustani *lūṭ*, plunder, as their own word for what they captured.
- It appeared in print in an Anglo-Indian vocabulary of 1788 and travelled with the army through the Indian and China campaigns.
- Colin Campbell used it in a letter of 1842, and Lord Elgin in 1858 (Yule and Burnell, 1886).

▷ Back

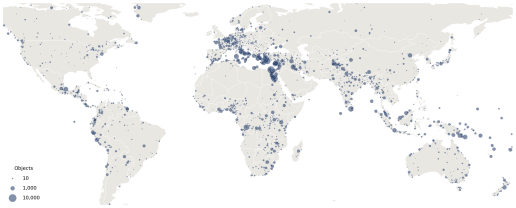
Campaign Cases, Channels and Timing

Campaign	Channel	What the catalog shows
Kerch, 1855	Officer donation	433 of 647 artifacts arrive in 1856, 225 from the army surgeon McPherson and 145 from Colonel Munro
Magdala, 1868	Army prize auction in the field	49 artifacts in 1868 against zero in each of the ten years before. Holmes bought some 350 manuscripts with a £1,000 brief
Kumasi, 1874	Trade auction, then state transfer	37 artifacts in 1874, 16 bought from Garrard's. 234 in 1896, 222 from the Gold Coast Government after the deposition of Prempeh I
Yuanmingyuan, 1860	Prize auction, then the salerooms	1,329 objects in 22 London auctions, 1861–97. Only 13 Beijing artifacts within four years, and the bulk from the 1880s on
Mandalay, 1885	Auction on the spot	Palace contents sold for about £60,000. 38 artifacts, mostly from political-service officers
Gyantse, 1904	Individual trophies	34 of 41 artifacts in 1905, after the Hague Convention of 1899. Most carry no acquisition source

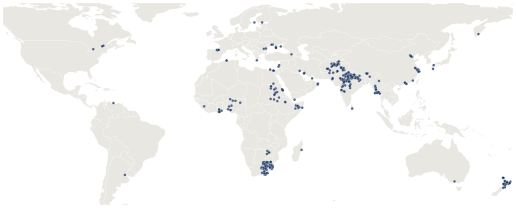
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Artifact Origins and British Battles

Artifact origins



British battles



▷ Back

Data Description (1/2)

1. Artifacts

	All	Year and location	Find spot, pre-1914	Long-run sample	Main sample
Number of Artifacts	1,532,147	1,218,797	496,231	367,836	139,657

Yearly Acquisitions	1753–1815	1816–1913	1914–2024
UK & Ireland	101.9	1,239.2	1,765.2
Rest of the World	238.9	3,819.1	5,237.8

Among artifacts in the main sample

Acquisition Method	Donations	Purchases	Others	Unrecorded
	63,678 (45.6%)	42,300 (30.3%)	20,179 (14.4%)	13,500 (9.7%)

Find-spot Precision	City or village	State/province	Country or broader
	91,767 (65.7%)	17,434 (12.5%)	30,456 (21.8%)

Location	Asia	Africa	Europe	Americas	Oceania
	48,766 (34.9%)	39,614 (28.4%)	24,438 (17.5%)	14,638 (10.5%)	12,201 (8.7%)

▷ next ▷ Back

Data Description (2/2)

2. Battles

Dataset coverage	All Battles	Battles 1816–1913	British battles 1816–1913
	8,401	2,118	474

Among British battles between 1816 and 1913

Location	Asia	Africa	Europe	Americas	Oceania
	240 (50.6%)	160 (33.8%)	25 (5.3%)	6 (1.3%)	43 (9.1%)

Battle outcomes	British won	British lost or tied
	365 (77.0%)	109 (23.0%)

3. Probate Records

Record counts	Total probate records	Years covered	Military officers
	16,149,806	1858–1996	13,556

4. Battle Leaders

Leader coverage	Total battle leaders	Within probate coverage	Matched to probate
	317	245	218

5. Rank Purchasing

Selected channels	By Purchase	Without Purchase	By Seniority	By Brevet
	24,070	12,431	287	2,315

▷ Back

Summary Statistics

	Mean	Median	SD	Min	Max	N
Panel A: Geographic Panels (1816–1913)						
City-Year Level						
Artifacts acquired	0.704	0.000	18.460	0	4204	198,450
Any British battle	0.002	0.000	0.040	0	1	198,450
Number of British battles	0.002	0.000	0.075	0	10	198,450
Grid-Cell-Year Level						
Artifacts acquired	0.655	0.000	17.778	0	4216	213,248
Any British battle	0.002	0.000	0.039	0	1	213,248
Number of British battles	0.002	0.000	0.073	0	17	213,248
Panel B: Battle Leader Level						
Probate value (1900 £)	20,421	6,317	35,374	0	282,468	218
Noble or royal descent	0.204	0.000	0.404	0	1	245
Father served in military	0.355	0.000	0.480	0	1	245
Panel C: Regiment-Year Purchase Panel (1848–1868)						
All promotions	7.480	6.000	6.093	0	51	1,176
Promotions by purchase	3.556	3.000	3.155	0	18	1,176

▷ Back

Why 1816–1913, and Why Not the UK and Ireland?

- The window runs from the end of the Napoleonic Wars to the First World War.
  - The Museum's collection only begins its steady growth in the nineteenth century.
  - The Napoleonic Wars and the First World War were mass continental conflicts, not the overseas expeditions the paper studies.
  - The Hague Conventions of 1899 and 1907 introduced international restrictions on wartime seizure.
- Artifacts from the UK and Ireland are excluded because the outcome is overseas cultural extraction.
- Extending the panel back to the Museum's founding in 1753, which adds the Seven Years' War, the American Revolution, and the French wars, gives similar estimates. ▷ go

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A Full Catalog Record

Object Type
bell

Museum number
Af1897,-.498

Description
Bell; lost-wax cast in brass. Quadrangular in form with slightly flaring sides and strap handle. Decorated on front panel with leopard's face in low relief with background of concentric arches and stippling. Side panels have angled interlace patterns with stippling, back panel has lozenge pattern with inset circular wheels. Each panel bordered by plaitwork. Iron...

View more

Production ethnic group
Made by: Edo

Production date
18thC (circa)

Production place
Made in: Benin City
Africa: sub-Saharan Africa: Nigeria: Edo State: Benin City

Excavator/field collector
Field Collection by: Major William Alexander Crawford Cockburn

Findspot
Found/Acquired: Benin City
Africa: sub-Saharan Africa: Nigeria: Edo State: Benin City

Materials
brass
iron

Technique
lost-wax cast

Dimensions
Height: 21.10 centimetres
Weight: 1528 grammes
Width: 11.20 centimetres
Depth: 12.80 centimetres

Inscriptions
Inscription type: inscription
Inscription position: label on back panel
Inscription language: English
Inscription content: 97-498. Benin City. C.F. 18 June 1897. Cockburn.
Inscription note: Registration number. Findspot. Christy Fund. Date of acquisition. Name of vendor.

Curator's comments
Read & Dalton 1899:
Bell of cast brass, square in form, with flat loop handle; each face has a design produced by means of a punch; the front has a scale pattern, with a leopard's face in relief in the middle. On the two sides is a trellis pattern, and on the back a lozenge design with rosettes; raised borders; iron clapper.

The workmanship and...

View more

Bibliographic references
Barley 2010 / The Art of Benin (p.65, fig. 43)
Read & Dalton 1899 / Antiquities from the City of Benin and from other parts of West Africa in the British Museum (pl VI.9)

Location
Not on display

Exhibition history
Exhibited:
1970-1973, 15 Dec - 14 Oct, London, Museum of Mankind, Divine Kingship in Africa
1991 Feb-Apr, Norwich, Sainsbury Centre for Visual Arts, Man and Metal in Ancient Nigeria
1993-1997 15 Dec - 31 Dec, London, Museum of Mankind, Great Benin: a West African Kingdom

Condition
Good; burnt in casting repairs on back panel of bell.

Subjects
wild cat/big cat

Acquisition name
Purchased from: Major William Alexander Crawford Cockburn

Acquisition notes
During the British expedition to Benin City (Edo) in 1897 objects made of brass, bronze, ivory, coral, and wood were looted by British soldiers from the royal palace, its storerooms, and compounds.

Some of these objects were sold or exchanged in West Africa. However, many were brought to the UK where they were retained by soldiers of the expedition and subsequently inherited by their families; put up for auction; or donated, lent, or sold to museums.

See Collection File: Af1897,-.498-563.

Department
Africa, Oceania and the Americas

Registration number
Af1897,-.498

Additional IDs
CDMS number: Af1897C3.498 (old CDMS no.)



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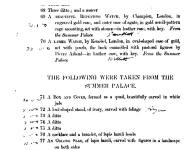
What the Catalog Measures

- Not a census of what British forces took. Many objects stayed in houses, messes, regimental and colonial museums, or went abroad.
- A large, systematic, lower-bound record of part of the flow.
  - A stable share of battle-induced flows makes the catalog a proxy for them.
  - Changes in the Museum's collecting or documentation are absorbed by year effects.
  - Places that generate more artifacts for any reason are absorbed by location effects.
- Disposal has long been barred. In 1753 the trustees told Parliament the collection "should be preserved entire", and in 1934 a Keeper stated that "The Trustees have no power to sell any object under their charge" (Caygill, 2002).

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Auction Sale Catalogues

Christie's, 14 July 1864



68 Three ditto; and a saucer

69 A magnificent Dessert Service, by Champion, London, in engraved gold vases, and outer case of agate, in gold mounted ormolu case mounting, set with stones—in leather case, with key. *From the Summer Palace.*

70 A Lemon Warmer, by Kendrick, London, in embossed case of gold, set with panels, the back enamelled with pastoral figures by Peyre Ainé—in leather case, with key. *From the Summer Palace.*

THE FOLLOWING WERE TAKEN FROM THE SUMMER PALACE.

71 A Box and Cover, formed as a goad, beautifully carved in white jade

72 A bas-relief stand, of ivory, carved with foliage

73 A ditto

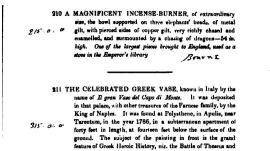
74 A ditto

75 A ditto

76 A Tablet and a bracelet, of lapis lazuli beads

77 An Oblong Stone, of lapis lazuli, carved with figures in a landscape on both sides

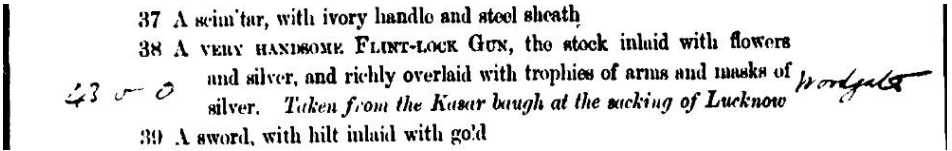
Christie's, 11 June 1863



210 A MAGNIFICENT INCENSE-BURNER, of extraordinary size, the bowl supported on three elephants' heads, of metal gilt, with pierced sides of copper gilt, very richly chased and enamelled, and surmounted by a chasing of dragons—94 in. high. *One of the largest pieces brought to England, and as a stone in the Emperor's library.*

211 THE CELEBRATED GREEK VASE, known in Italy by the name of *Il gran Vase del Capo di Monte*. It was disposited in that palace, with other treasures of the Farnese family, by the King of Naples. It was found at Pulysthenes, in Apulia, near Tarentum, in the year 1785, in a subterranean apartment of forty feet in length, at fourteen feet below the surface of the ground. The subject of the painting in front is the grand feature of Greek Heroic History, viz. the Battle of Thessa and

Christie's, 16 February 1865



37 A scim'tar, with ivory handle and steel sheath

38 A VERY HANDSOME FLINT-LOCK GUN, the stock inlaid with flowers and silver, and richly overlaid with trophies of arms and masks of silver. *Taken from the Kasar baugh at the sacking of Lucknow*

43 10 0

39 A sword, with hilt inlaid with gold

▷ coverage and prices ▷ data ▷ Back

How Complete Is the Catalog?

- The Museum put about 2 million records online in 2024, of 8 million in the whole collection. The scrape returns 1,532,147 records through every geographic term, so artifacts without a place are outside it and could not enter the design anyway.
- Against the Museum's internal registration counts by decade (MacDonald, 2023), the scrape covers a stable fraction and tracks the decade-to-decade movements ($\rho = 0.90$).
  - Proportional under-coverage attenuates nothing in a within-unit design. Selectively timed digitization would, and the co-movement rules it out.
- Later additions to the online catalog add artifacts but do not alter the acquisition years or find spots already recorded.
- Entry channels, 1816–1913: 3% government transfer, 1% gifts and 1% sales from military sources, 16% dealers and auction houses, 63% private persons. The dealer share is a lower bound.

▷ Back

Geocoding and Its Audit

- Find spots range from “Summer Palace (Beijing)” to “India (historic—pre-1947) (North-west)”. Standard geocoders mishandle Oudh, Gold Coast, or Punjab (India).
- A language model reads each place with its full parent chain and returns coordinates and a precision class. 1,532,147 records reduce to 53,176 unique place entries.
- Precision in the main sample: 65.7% of artifacts at a city or village, 12.5% at a state or province, 21.8% at a country.
- Random audit of 100 artifacts: mean error 5.5 km, median 1.0 km. 96 within 20 km, two within 50 km, one within 100 km, and one mission station in Malawi 206 km off.
- Every place description attached to more than ten artifacts, 8,698 descriptions and 88% of artifact–place links, was checked by hand.
- Precise-only, grid, and proportional-distribution designs test the imprecise class. ▷ go

▷ Back

Battles and Commanders

Benin ❙ 1897 ❙ British Conquest of Nigeria

Avenging the massacre at **Ugbine** (4 January), a 1,200-strong British Punitive Expedition under Admiral Harry Rawson converged on King Oba at Benin City in modern Nigeria, whose army was destroyed by Maxim guns and artillery. King Oba's capital was razed and he was exiled. His son-in-law Ologbosheri was eventually captured and executed for the attack at Ugbine (18 February 1897).

- I digitize Jaques' *Dictionary of Battles and Sieges*, 8,401 entries.
- A language model extracts belligerents, leaders, place, year, and outcome. I check every field of the 474 British battles of 1816–1913 by hand.
- A British battle: a state-backed British force fought, including the Indian Army. Private hires are out.
- I research the 317 commanders by hand: ranks, honors, and family background.

▷ coding rule and HCED ▷ other battle sources

▷ dating long battles

▷ leaders ▷ Back

Coding British Battles, and HCED

- British: a Crown force or an explicitly Crown-backed force fought as a combatant under British command, including the East India Company and Indian Army, chartered companies, warranted volunteer corps, and colonial corps. Out: privately employed officers (Cochrane in Greece) and warships present but not engaged (Valparaíso).
- Every field of the 474 British battles is checked by hand, and the dictionary's own errors (ranks overstated, names garbled) are corrected with a source. A random audit of 100 non-British battles: 98.8% of 500 judgments correct, and 98 of 99 coordinates within 50 km.
- HCED (Miller and Bakar, 2023) structures the same dictionary: 502 British engagements against my 474, with 443 matching. The HCED-only cases are the Riel Rebellion militia actions, Cochrane's Greek command, and American Civil War engagements.
- Wikidata carries British-participant statements for only 195 battles of 1816–1913. Battle honors and medal clasps are selected on success.

▷ Back

Probate Digitization and Linkage

- 16.1 million grants over 139 calendar years, read from the scanned calendars with a vision-language model and parsed by a language model into name, value, address, occupation (to 1907), and executor.
- Audit of 20 random pages, 431 records: every printed entry has a record, and 98.1% of values fall within 10% of the printed sum. Three further rounds give 89 to 98%. Early entries state ceilings ("Effects under £1,500").
- Every commander's value is read by hand from the original page and deflated to 1900 prices with the Retail Prices Index.
- Linkage: 59 of the 317 commanders died before 1858 and 13 lack a confirmed death year or coverage, leaving 245. I match 218 (89%). Of the 27 unmatched, 13 lived and died overseas and 14 left no grant under their name.
- The 13,556 *Officers in probate* come from the 1858–1906 occupation field, with civilian uses ("General Manager") dropped and every commander removed.

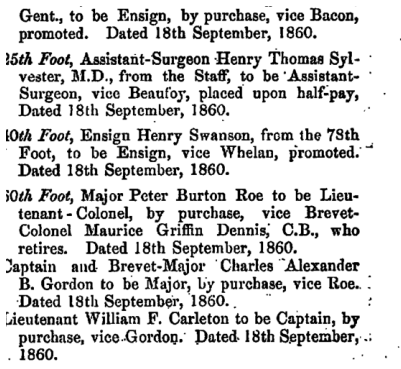
▷ Back

Gazette Extraction and Its Audit

- 27,737 issues and 656,357 pages, 1665–1914. For 1816–1870, 6,531 of 6,597 issue numbers are available.
- A vision-language model converts pages to text, and a language model reads the 21,929 officer-section blocks into one row per appointment, promotion, retirement, transfer, or exchange, each with a quotation of the source text.
- 274,485 events for 110,378 officer-name strings. 102,368 promotions, of which 36,501 state a channel and 24,070 (66%) were by purchase, the split Bruce (1980) reports.
- Audit of eighteen issues, 1,059 rows: name and ranks correct in 98.7%, the purchase channel in 994 of 995. Recall: 98.5% of 1,010 independently enumerated events.
- No by-purchase notice appears in the navy, militia, or volunteer sections, where commissions were not sold.
- The analysis counts by-purchase promotions by regiment and year, so only the purchase-channel field can bias it.

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From Gazette Notice to Structured Record



Gent., to be Ensign, by purchase, vice Bacon, promoted. Dated 18th September, 1860.

25*th Foot*, Assistant-Surgeon Henry Thomas Sylvester, M.D., from the Staff, to be Assistant-Surgeon, vice Beaufoy, placed upon half-pay, Dated 18th September, 1860.

40*th Foot*, Ensign Henry Swanson, from the 78th Foot, to be Ensign, vice Whelan, promoted. Dated 18th September, 1860.

60*th Foot*, Major Peter Burton Roe to be Lieutenant-Colonel, by purchase, vice Brevet-Colonel Maurice Griffin Dennis, C.B., who retires. Dated 18th September, 1860.

Captain and Brevet-Major Charles Alexander B. Gordon to be Major, by purchase, vice Roe. Dated 18th September, 1860.

Lieutenant William F. Carleton to be Captain, by purchase, vice Gordon. Dated 18th September, 1860.

Field	Extracted value
Officer	Peter Burton Roe
Rank	Major → Lieutenant-Colonel
Regiment	60th Foot
Channel	**by purchase**
Replaces (*vice*)	Maurice Griffin Dennis
Date	18 September 1860

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Auction Catalogues, Coverage and Prices

- Lugt's *Répertoire* lists 57,436 sales for 1750–1900. Brill scanned 27,870 of them, and I obtained 27,751: 32,568 copies, 900,279 pages, 99.3% readable.
- Against the union catalogue of research libraries, the corpus holds 46% of Christie's London sales, 53% of Phillips, 20% of Foster's, 13% of Sotheby's, and 67% of the Hôtel Drouot.
- Campaign lots: a regular-expression pass over 14.5 million lots flags 14,484 candidates, and language-model readers keep 1,803 lots in 138 sales, 95 of them held in Britain. Lots that merely depict a war, use a place as a style name, or name a non-British taker are excluded.
- Prices: 83 legible hammer prices, median £5 10s, total £2,869 (US$650,000 at 2025 prices). The largest, a pair of gold Yuanmingyuan vases sold in Paris in 1872, about 25,000 francs.
- Limits: the corpus ends in 1900, and the counts measure objects marketed on campaign provenance, which sellers had an incentive to advertise.

▷ data ▷ Back

Why PPML?

- The outcome is a skewed count with a large mass at zero.
- PPML gives correct marginal effects for such outcomes (Mullahy and Norton, 2024). Effects on log or asinh outcomes depend on the units when the treatment moves the extensive margin (Chen and Roth, 2024).
- Poisson is consistent for the conditional mean whatever the distribution of the count, so the zeros cost efficiency, not consistency (Gourieroux et al., 1984; Santos Silva and Tenreyro, 2006).
- OLS in levels, logs, and asinh, and a negative binomial give positive estimates. ▷ go
- On the extensive margin, a battle raises the chance of any acquisition by 4.1 pp on a 4.6% mean. ▷ go

▷ Back

Why Not Sun–Abraham or Borusyak et al. in the Short Run?

- The battle treatment is recurring and non-absorbing: 73 of the 219 battle cities saw British fighting in more than one year, and the indicator returns to zero once the window passes.
- Sun and Abraham (2021) and Borusyak et al. (2024) assume a unit stays treated after its first treatment. de Chaisemartin and D'Haultfœuille (2026) allows switching off but measures effects from a city's first battle.
- So the short run uses two designs that keep the recurring treatment and compare each battle only with cities without a battle in the surrounding years: stacked PPML on clean sub-experiments (Cengiz et al., 2019) and local-projections DiD (Dube et al., 2025). ▷ Table 3 ▷ the stacking
- The long run treats a city's first battle as absorbing at the decade level, where those estimators apply: Borusyak et al., Sun–Abraham, de Chaisemartin–D'Haultfœuille, and synthetic DiD (Arkhangelsky et al., 2021). ▷ go

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The Comparison Behind the Artifact Estimates

- Baseline: compare acquisition changes within a city with changes in other sample cities in the same years.
- Sample: cities with any artifact acquisition or British battle, 1816–1913.
- Strictest city specification: compare with similarly sized cities in the same country and year. Country and size-group trends can differ.

Identifying assumption

Conditional on the fixed effects, the timing of a battle within a place was unrelated to the acquisitions the Museum would otherwise have made from it.

A local shock that both triggered fighting and moved objects would violate this assumption.

▷ tests of this concern ▷ comparisons without recent battles ▷ Back

Stacked and Local-Projection Estimates

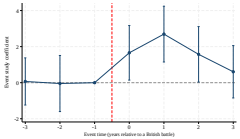
Comparisons with recently treated units do not drive the positive baseline effect.

	Number of Artifacts Acquired This Year					
	(1)	(2)	(3)	(4)	(5)	(6)
Previous Year Had Any British Battle	2.150***	2.694***	0.245***	2.413***	3.220***	0.204***
	(0.534)	(0.789)	(0.088)	(0.605)	(0.837)	(0.078)
Unit of observation	City	City	City	1° Grid	1° Grid	1° Grid
Estimation	PPML	PPML (Stacked)	LP–DiD (asinh)	PPML	PPML (Stacked)	LP–DiD (asinh)
City or Grid FE	Yes	Yes	Differenced	Yes	Yes	Differenced
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
FE × Stack	No	Yes	No	No	Yes	No
Number of Observations	198,450	935,935	180,937	213,248	1,006,096	198,509
Mean of dep. var.	0.704	0.697	0.091	0.655	0.638	0.087
SD of dep. var.	18.460	19.496	0.501	17.778	18.549	0.485

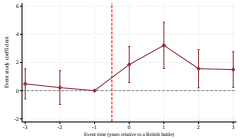
▷ stacked design ▷ event studies ▷ comparison groups ▷ Back

Event Studies, Stacked and Local Projections

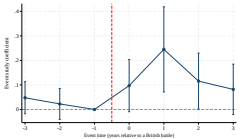
Stacked PPML, city



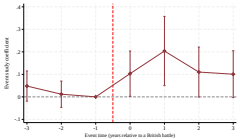
Stacked PPML, grid



Local projections, city



Local projections, grid



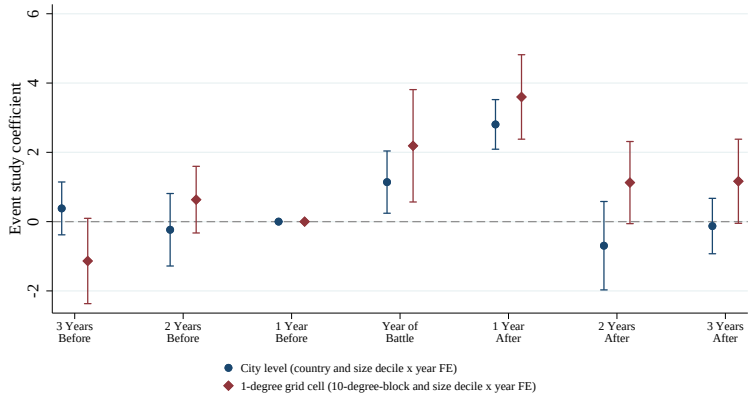
▷ the stacking ▷ Back

The Stacked Design

- A sub-experiment around every British battle year b: event years $b-3$ to $b+3$, the units cleanly treated in b, and controls.
- Clean: no other British battle at the unit in $[b-6, b+3]$. Controls: units with no British battle in the same range. Units treated in the distant past or future may serve as controls once their dynamics have cleared.
- Estimation absorbs unit-by-stack and year-by-stack effects, so no comparison runs across sub-experiments. A unit appears in many stacks: 935,935 city-year rows against 198,450 in the baseline, each copy inside one cluster.
- Restricting controls to units that never host a British battle moves the one-year coefficient from 2.69 to 2.71 (city) and 3.22 to 3.25 (grid).
- Dropping any unit with a battle of another European power, or any battle in Europe, removes four of 201 episodes and leaves every coefficient unchanged to the second decimal.

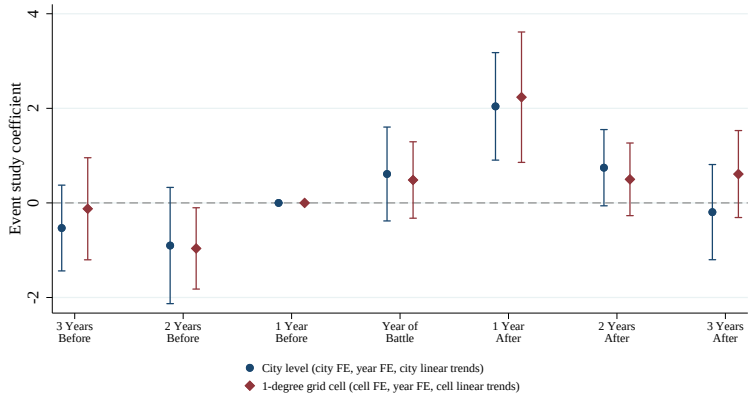
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Event Study with Geographic and Size-by-Year Effects



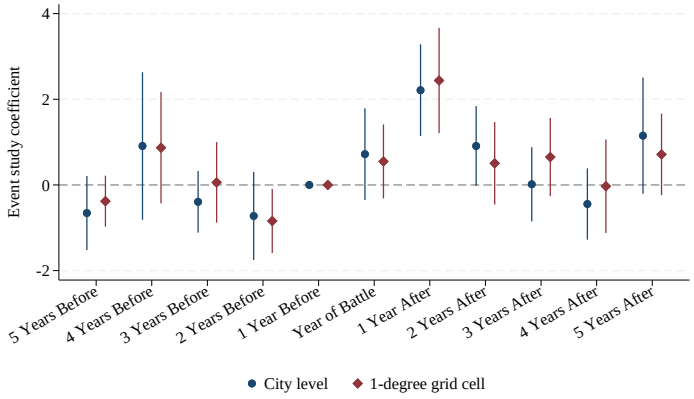
▷ Back

Event Study with Unit-Specific Trends



▷ Back

Event Study, ±5 Years



▷ Back

Functional Form

	Number of Artifacts Acquired This Year				
	(1)	(2)	(3)	(4)	(5)
Panel A. City					
Previous Year Had Any British Battle	2.150***	4.358**	1.009*	0.160***	2.756***
	(0.534)	(2.214)	(0.536)	(0.059)	(0.474)
Estimation	PPML	OLS	OLS	OLS	Negative binomial
Dependent variable	Count	Count	ln(count), positive	asinh(count)	Count
City or Grid FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Number of Observations	198,450	194,432	7,835	194,432	198,450
Mean of dep. var.	0.704	0.681	1.242	0.089	0.704
SD of dep. var.	18.460	18.183	1.385	0.493	18.460
Panel B. 1° Grid					
Previous Year Had Any British Battle	2.413***	7.235*	0.956*	0.144***	2.677***
	(0.605)	(3.951)	(0.513)	(0.051)	(0.513)
Estimation	PPML	OLS	OLS	OLS	Negative binomial
Dependent variable	Count	Count	ln(count), positive	asinh(count)	Count
City or Grid FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Number of Observations	213,248	213,248	8,476	213,248	213,248
Mean of dep. var.	0.655	0.655	1.215	0.087	0.655
SD of dep. var.	17.778	17.778	1.372	0.485	17.778

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Extensive Margin

	Any Artifact Acquired This Year (=1)			
	(1)	(2)	(3)	(4)
Previous Year Had Any British Battle	0.0413***	0.0448***	0.0358***	0.0394***
	(0.0135)	(0.0141)	(0.0122)	(0.0124)
Unit of observation	City	City	1° Grid	1° Grid
City or Grid FE	Yes	Yes	Yes	Yes
Year FE	Yes		Yes	
Country × Year FE	No	Yes		
Population Decile × Year FE	No	Yes		
Area Decile × Year FE	No	Yes		
10° Block × Year FE			No	Yes
Cell Pop./Settlement Decile × Year FE			No	Yes
Number of Observations	198,450	198,450	213,248	213,248
Mean of dep. var.	0.046	0.046	0.044	0.044

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Alternative Geocoding

	Number of Artifacts Acquired This Year			
	(1)	(2)	(3)	(4)
Artifact-location measure:	Baseline	Proportional distribution	City/village-precise find spot	Find spot with production-place fallback
Panel A: City				
Previous Year Had Any British Battle	2.150***	2.273***	2.671***	1.425**
	(0.534)	(0.495)	(0.569)	(0.591)
Number of Observations	198,450	198,450	198,450	227,066
Mean of dep. var.	0.704	0.704	0.462	1.424
SD of dep. var.	18.460	17.976	16.645	22.496
Panel B: 1° Grid				
Previous Year Had Any British Battle	2.413***	2.657***	3.017***	1.704***
	(0.605)	(0.556)	(0.647)	(0.595)
Number of Observations	213,248	213,248	213,248	246,176
Mean of dep. var.	0.655	0.655	0.430	1.353
SD of dep. var.	17.778	17.214	16.067	22.107
Artifacts in sample	139,657	139,657	91,767	332,968
City or Grid FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

▷ data ▷ Back

Exhibited Artifacts and Personal Donations

Outcome:	Number of Artifacts Acquired This Year					
	(1) All artifacts	(2) Ever exhibited	(3) Excluding donations	(4) All artifacts	(5) Ever exhibited	(6) Excluding donations
Previous Year Had Any British Battle	2.150***	3.270***	2.104***	2.413***	3.131***	2.472***
	(0.534)	(0.517)	(0.649)	(0.605)	(0.514)	(0.763)
Unit of observation	City	City	City	1° Grid	1° Grid	1° Grid
City or Grid FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Number of Observations	184,494	67,022	114,760	201,081	73,226	125,970
Mean of dep. var.	0.704	0.043	0.484	0.655	0.040	0.450
SD of dep. var.	18.460	1.172	17.604	17.778	1.125	17.035
Artifacts in outcome	139,657	8,616	95,971	139,657	8,616	95,971

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Alternative Battle Sources

	Number of Artifacts Acquired This Year			
Battle definition:	(1) Baseline (audited)	(2) Regular forces only	(3) HCED British tag	(4) Wikidata British battles
Panel A: City				
Previous Year Had Any British Battle	2.150***	2.159***	2.123***	1.739***
	(0.534)	(0.542)	(0.531)	(0.651)
Number of Observations	198,450	198,450	198,450	198,450
Mean of dep. var.	0.704	0.704	0.704	0.704
SD of dep. var.	18.460	18.460	18.460	18.460
Panel B: 1° Grid				
Previous Year Had Any British Battle	2.413***	2.416***	2.399***	2.499***
	(0.605)	(0.606)	(0.604)	(0.862)
Number of Observations	213,248	213,248	213,248	213,248
Mean of dep. var.	0.655	0.655	0.655	0.655
SD of dep. var.	17.778	17.778	17.778	17.778
Battles entering treatment	474	439	494	143
City or Grid FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

▷ data ▷ Back

Dating Multi-Year Battles

	Number of Artifacts Acquired This Year					
	(1)	(2)	(3)	(4)	(5)	(6)
Previous Year Had Any British Battle	2.150***	2.117***	2.117***	2.413***	2.387***	2.387***
	(0.534)	(0.530)	(0.530)	(0.605)	(0.605)	(0.605)
Unit of observation	City	City	City	1° Grid	1° Grid	1° Grid
Battle Year Assignment	Start Year	End Year	Random Year	Start Year	End Year	Random Year
City or Grid FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Number of Observations	198,450	198,450	198,450	213,248	213,248	213,248
Mean of dep. var.	0.704	0.704	0.704	0.655	0.655	0.655
SD of dep. var.	18.460	18.460	18.460	17.778	17.778	17.778

▷ data ▷ Back

Influential Observations (1/2)

	Number of Artifacts Acquired This Year			
Panel A: Winsorizing and trimming				
	(1)	(2)	(3)	
Previous Year Had Any British Battle	2.150***	2.137***	1.886***	
	(0.534)	(0.479)	(0.493)	
Outlier Handling	None	Winsor at event p99	Trim at event p99	
Number of Observations	198,450	198,450	198,359	
Mean of dep. var.	0.704	0.528	0.425	
SD of dep. var.	18.460	6.938	5.003	
Panel B: Top-coding the outcome				
	(1)	(2)	(3)	
Previous Year Had Any British Battle	0.976***	1.474***	1.785***	
	(0.284)	(0.392)	(0.436)	
Top-Coding	Capped at 10	Capped at 50	Capped at 100	
Number of Observations	198,450	198,450	198,450	
Mean of dep. var.	0.186	0.360	0.441	
SD of dep. var.	1.142	3.184	4.667	
Panel C: Dropping top artifact cities				
	(1)	(2)	(3)	(4)
Previous Year Had Any British Battle	2.101***	2.064***	2.042***	2.189***
	(0.532)	(0.536)	(0.542)	(0.545)
Sample	Drop Top 1	Drop Top 3	Drop Top 5	Drop Top 10
Number of Observations	198,352	198,156	197,960	197,470
Mean of dep. var.	0.670	0.619	0.584	0.525
SD of dep. var.	15.536	14.056	13.124	10.347

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Influential Observations (2/2)

Panel D: Dropping top battle cities				
	(1)	(2)	(3)	(4)
Previous Year Had Any British Battle	2.150***	2.273***	2.276***	2.040***
	(0.534)	(0.533)	(0.533)	(0.605)
Sample	Drop Top 1	Drop Top 3	Drop Top 5	Drop Top 10
Number of Observations	198,352	198,156	197,960	197,470
Mean of dep. var.	0.704	0.703	0.704	0.703
SD of dep. var.	18.465	18.470	18.479	18.494
Panel E: Country jackknife				
	(1)	(2)	(3)	(4)
Previous Year Had Any British Battle	2.223***	2.321***	1.558***	2.046***
	(0.547)	(0.526)	(0.580)	(0.591)
Country Dropped	India	China	Afghanistan	Nigeria
Number of Observations	187,180	193,550	196,882	195,020
Mean of dep. var.	0.723	0.707	0.698	0.711
SD of dep. var.	18.965	18.635	18.360	18.614
City FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

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Sample, Spatial Units, and Inference (1/2)

	Number of Artifacts Acquired This Year				
Panel A: Sample support					
	(1)	(2)	(3)	(4)	(5)
Sample	Battle or artifact	All cities	Battle cities	Artifact cities	Battle and artifact
Previous Year Had Any British Battle	2.150***	2.150***	2.545***	2.150***	2.545***
	(0.534)	(0.534)	(0.486)	(0.534)	(0.486)
Number of Observations	198,450	719,516	21,462	190,218	13,230
Mean of dep. var.	0.704	0.194	0.403	0.734	0.654
Panel B: Cities documented as existing					
	(1)	(2)	(3)		
Restriction	Baseline	Existed by 1900	Existed by 1815		
Previous Year Had Any British Battle	2.150***	2.212***	2.723***		
	(0.534)	(0.553)	(0.607)		
Number of Observations	198,450	161,504	120,540		
Mean of dep. var.	0.704	0.759	0.860		
Number of cities	2,025	1,648	1,230		
Panel C: Panel extended to 1753–1913					
	(1)	(2)			
Unit of observation	City	1° Grid			
Previous Year Had Any British Battle	2.134***	2.394***			
	(0.526)	(0.596)			
Number of Observations	350,658	373,037			
Mean of dep. var.	0.403	0.379			

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Sample, Spatial Units, and Inference (2/2)

Panel D: Coarser spatial units

	(1)	(2)	(3)	(4)	(5)
Unit of observation	City	State/Prov.	1° Grid	2° Grid	5° Grid
Previous Year Had Any British Battle	2.150***	1.832***	2.413***	1.789***	1.182**
	(0.534)	(0.474)	(0.605)	(0.679)	(0.569)
Number of units	2,025	1,198	2,176	1,335	572
Number of Observations	198,450	117,404	213,248	130,830	56,056
Mean of dep. var.	0.704	1.190	0.655	1.067	2.491

Panel E: Conley standard errors, city panel, inverse hyperbolic sine of the count

	(1)	(2)	(3)	(4)
Conley cutoff	100 km	200 km	500 km	1,000 km
Previous Year Had Any British Battle	0.159***	0.159***	0.159***	0.159**
	(0.051)	(0.054)	(0.059)	(0.062)
Number of Observations	198,450	198,450	198,450	198,450
Mean of dep. var.	0.091	0.091	0.091	0.091

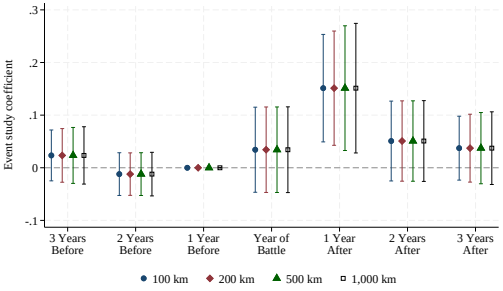
Panel F: Conley standard errors, 1° grid panel, inverse hyperbolic sine of the count

	(1)	(2)	(3)	(4)
Conley cutoff	100 km	200 km	500 km	1,000 km
Previous Year Had Any British Battle	0.144***	0.144***	0.144**	0.144**
	(0.051)	(0.056)	(0.060)	(0.063)
Number of Observations	213,248	213,248	213,248	213,248
Mean of dep. var.	0.087	0.087	0.087	0.087
Unit FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

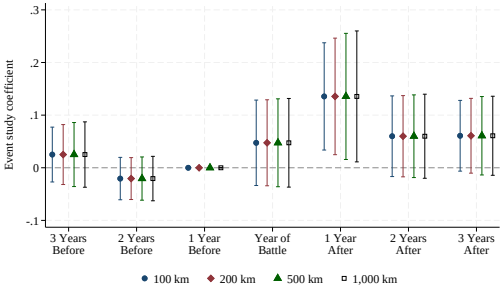
▷ Back

Conley Standard Errors

City panel



1° grid panel



▷ Back

Unit-Specific Linear Trends

	Number of Artifacts Acquired This Year			
	(1)	(2)	(3)	(4)
Previous Year Had Any British Battle	2.150***	2.017***	2.413***	2.231***
	(0.534)	(0.552)	(0.605)	(0.668)
Unit of observation	City	City	1° Grid	1° Grid
City or Grid FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Unit-specific linear trend	No	Yes	No	Yes
Number of Observations	198,450	198,450	213,248	213,248
Mean of dep. var.	0.704	0.704	0.655	0.655
SD of dep. var.	18.460	18.460	17.778	17.778

▷ Back

Do Prior Acquisitions Predict a British Battle?

	British Battle This Year (× 100)					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Regressor is any acquisition in the previous year (a separate regression in each column)						
Any Acquisition in the Previous Year	-0.053	-0.042	-0.027	-0.047	-0.050	-0.039
	(0.039)	(0.041)	(0.042)	(0.038)	(0.041)	(0.041)
Number of Observations	192,448	189,247	189,053	211,072	206,513	206,513
Mean of dep. var.	0.161	0.161	0.161	0.152	0.155	0.155
Panel B: Regressor is any acquisition in any of the previous three years (a separate regression in each column)						
Any Acquisition in the Previous Three Years	-0.025	0.006	0.009	-0.024	-0.004	0.005
	(0.028)	(0.028)	(0.029)	(0.027)	(0.028)	(0.029)
Number of Observations	188,480	185,345	185,155	206,720	202,255	202,255
Mean of dep. var.	0.158	0.158	0.158	0.150	0.153	0.153
Unit of observation	City	City	City	1° Grid	1° Grid	1° Grid
City or Grid FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	–	–	Yes	–	–
Country × Year FE	No	Yes	Yes	–	–	–
Pop./Area Decile × Year FE	No	No	Yes	–	–	–
10° Block × Year FE	–	–	–	No	Yes	Yes
Cell Pop./Settlement Decile × Year FE	–	–	–	No	No	Yes

▷ Back

Did Britain Fight for the Spoils?

- The assumption is not that battles were random, but that the timing of a battle within a place was unrelated to the acquisitions the Museum would otherwise have made from it.
- City effects absorb that Britain selected its targets, including for what they promised to yield.
- An expedition timed toward a place whose acquisitions were already rising would show up as a pre-battle trend. The event studies show none, and recent acquisitions do not predict a battle.
- Where the other side set the timing, in the 34 incident-triggered battles, acquisitions still rise.

▷ Back

Why Not an Instrument?

- The incident coding supports a robustness split, not an instrumental-variables design.
  - An incident can affect acquisitions through public attention as well as through the battle.
  - Some expeditions were prepared before the incident that occasioned them: Benin was planned some four and a half years in advance (Hicks, 2020). The 1839 Aden expedition is excluded for the same reason.
- Coding rule: an expedition set off by a discrete, datable anti-British act that preceded any British decision to fight. 34 of 474 battles, 21 at high confidence, checked by hand against the historical record.
- Examples: Theodore's imprisonment of British envoys before Magdala, the seizure of the *Arrow*'s crew before the Second Opium War.
- The split shows that when the timing was forced from outside, the effect is similarly strong, if not stronger.

▷ the 34 battles ▷ Back

The 34 Incident-Triggered Battles (1/2)

Campaign	Battles	Trigger
Corsair Wars	Algiers 1816, 1824	corsair attacks on shipping and Christian slavery
Third Maratha War	Kirkee 1817	the Peshwa burned the Residency at Poona
Anglo-Arab Wars	Ras al-Khaimah 1819, Balad Bani Bu Ali 1821	piracy in the Gulf, then a defeat at Sur to avenge
Bharatpur	Bharatpur 1825–26	a usurper seized the throne from the British-recognized heir
First Opium War	Chuanbi 1839	an attack on British merchants off Kowloon
First Afghan War	Kalat 1839; Jagdalak, Maidan, Tezin 1842	hindering the expedition, then the destruction of the Kabul garrison
First Sikh War	Baddowal, Aliwal 1846	the Sikh army crossed the Sutlej into British Punjab
Crimean War	Odessa 1854	guns fired on a frigate under a flag of truce
Second Opium War	Fatshan Creek, Guangzhou 1857	men seized off the *Arrow*
Anglo-Persian War	Khoosh-Ab 1857	Persia's capture of Herat

▷ next ▷ Back

The 34 Incident-Triggered Battles (2/2)

Campaign	Battles	Trigger
Satsuma	Kagoshima 1863	the merchant Richardson killed
Second New Zealand War	Katikara 1863	a nine-man escort ambushed and killed
Shimonoseki	Shimonoseki 1864	batteries fired on merchant shipping
Ethiopia	Arogi, Magdala 1868	the envoy and other Europeans imprisoned
Second Afghan War	Charasia, Sherpur 1879	the envoy Cavagnari murdered
Nigeria	Benin 1897	the massacre at Ugbine
Sudan	Omdurman 1898	the death of Gordon at Khartoum
Boxer Rebellion	Langfang, Dagu Forts, Tianjin (two), Beicang, Yancun, Beijing 1900	the German minister murdered and the legations besieged
Northern Nigeria	Kano 1903	the Resident Moloney murdered at Keffi

Thirty are reprisals for a specific injury and four enforce trade and shipping. 21 of the 34 are flagged at high confidence. The 1839 Aden expedition is excluded as the seizure of a long-sought coaling station.

▷ Back

Civil-War and French-Only Battles

	Number of Artifacts Acquired This Year					
	(1)	(2)	(3)	(4)	(5)	(6)
Previous Year Had Any British Battle	2.150***			2.413***		
	(0.534)			(0.605)		
Previous Year Had Civil-War Battle		-1.455*			-1.366*	
		(0.878)			(0.729)	
Previous Year Had French-Only Battle			-2.225***			-2.460***
			(0.491)			(0.436)
Unit of observation	City	City	City	1° Grid	1° Grid	1° Grid
City or Grid FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Number of Observations	198,450	198,450	198,450	213,248	213,248	213,248
Mean of dep. var.	0.704	0.704	0.704	0.655	0.655	0.655
SD of dep. var.	18.460	18.460	18.460	17.778	17.778	17.778

Civil war: both belligerents are domestic parties of one polity and no foreign force took part, which holds for 642 of the 2,118 in-window battles, 597 at high confidence. Taiping engagements fought beside the Ever Victorious Army count as foreign-assisted.

▷ Back

Long-Run Comparison and Delayed Arrivals

- Exposure begins with the decade of a city's first British battle in 1816–1913. Later battles do not reset the clock.
- Controls: never-battle cities, plus cities before their first battle where permitted. City and decade effects absorb permanent differences and common collecting changes.

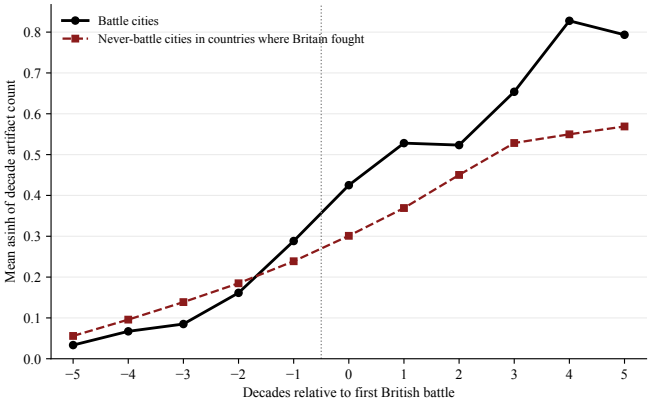
Identifying assumption

Without the battle, the transformed acquisition counts of exposed and control cities would have followed the same trends.

- Fast arrivals: prize sales and state transfers. Slow arrivals: officers' collections, dealers, and estates.
- Raw counts: 2.8 artifacts per battle city in the year after its first battle, and 31 over the five decades after the battle decade.

▷ raw comparison ▷ controls by estimator ▷ arrival lags ▷ Back

Battle Cities and Cities with No British Battle



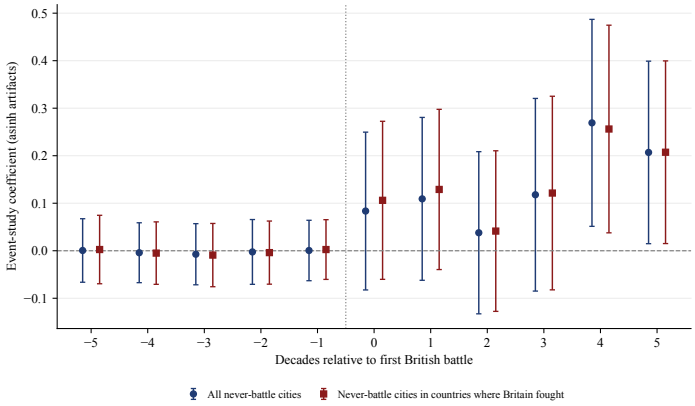
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Pooled Long-Run Estimates

	Asinh of decade artifact count					Any acquisition
	Staggered event study			Synthetic DiD		Staggered event study
	(1)	(2)	(3)	(4)	(5)	(6)
Post-battle average, decades 0–5	0.172***	0.131***	0.112*	0.137**	0.144**	0.039***
	(0.056)	(0.044)	(0.057)	(0.060)	(0.061)	(0.014)
Estimator	BJS	SA	dCDH	SDID	SDID	SA
Control group	No or not-yet British battle	No British battle	No or not-yet British battle	No British battle	No British battle, battle countries	No British battle
City FE	Yes	Yes	Yes	Yes	Yes	Yes
Decade FE	Yes	Yes	Yes	Yes	Yes	Yes
Number of Battle Cities	216	216	216	216	216	216
Number of Control Cities	2,883	2,883	2,883	2,883	1,181	2,883
Number of Observations	84,320	86,772	86,772	63,135	27,393	86,772

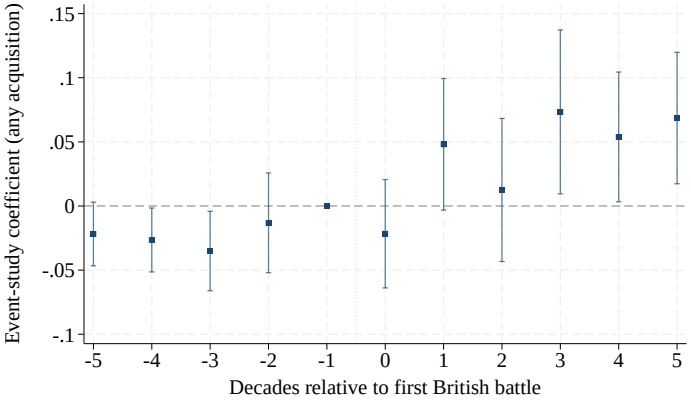
▷ any acquisition ▷ Back

Synthetic Difference-in-Differences



▷ Back

Any Acquisition, by Decade



▷ Back

Mechanisms: Measures

- British rule: I locate cities within British territories in the Cliopatria historical maps and verify the dates of territorial administration (Bennett et al., 2025).
- Military institutions: V-Dem measures whether non-conscript armed-forces personnel receive regular salaries from state coffers. I average this measure over 1800–1900 (Coppedge et al., 2026).
- Material culture: the technology-adoption index in 1500 averages agriculture, communications, transportation, industry, and military technology (Comin et al., 2010).

I split the two country-level measures at their medians, giving each matched country equal weight.

▷ Back

Military Institutions and Material Culture

	Military Professionalization (1800–1900)		Material Culture (1500 AD)	
	(1)	(2)	(3)	(4)
	Below Median	At/Above Median	Below Median	At/Above Median
Previous Year British Battle	3.984***	0.511	0.264	2.351***
	(0.642)	(0.652)	(0.558)	(0.598)
Country Median Cutoff	-0.336	-0.336	0.383	0.383
City FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Number of Countries	56	56	57	58
Number of Observations	58,604	92,530	63,976	93,605
Mean of Dep. Var.	0.929	0.636	0.428	0.952

▷ the measures ▷ Back

The Two Capacity Measures

- Military institutions: the V-Dem indicator of whether armed-forces personnel are salaried from state coffers, averaged over 1800–1900 (Coppedge et al., 2026). Material culture: the technology-adoption index of Comin et al. (2010) in 1500 AD.
- Each is split at its median across matched countries, one observation per country: 56 and 56 countries for institutions, 57 and 58 for material culture.
- Battle coefficient: 3.98 below the median of military professionalization and 0.51 above it. 2.35 above the median of material culture and 0.26 below it.
- The two splits are close to independent (correlation 0.08), and professionalization is only weakly related to income (0.19 with Maddison GDP per capita).
- Limits: 34 of the 112 countries with a V-Dem value host British battles, 42 countries rest on a single coded year, and Sudan, Nigeria, and Ghana have no nineteenth-century value and drop out.

▷ Back

The British-Rule Indicator

- A city-year is under British rule when its coordinates fall inside a British polygon of the Cliopatria polity database that year (Bennett et al., 2025), with a 10 km coastal buffer.
- Rule: ground Britain administered counts, whatever its legal form. Jurisdiction over British subjects inside a sovereign state does not.
- A language model flags suspicious city-years, flips, short spells, and holes, and I check each by hand against the dates of annexation and occupation.
- Added: the concessions in China, the Japanese treaty settlements, the Pacific protectorates, and seven cities no polygon reaches (Toronto, Nassau, Corfu, and others). Removed: Kabul in 1840–41, Sevastopol in 1856, the capitulations, and cities reached only across a border.
- Dates corrected: Egypt from 1882, Benin City from 1897, Natal from 1843. Gaps of at most two years are filled, which closes Cliopatria's withdrawal of the Raj polygon during the Mutiny for 51 cities.

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Acquisition Notes that Record a Taking

Event	Museum no.	Acq. year	Acquisition note (excerpt)
Sack of Benin City, 1897 (311)	Af1898,0115.10	1898	Following the British occupation of Benin City (Edo) in 1897 objects made of brass, ivory and wood were looted by British forces from the royal palace, its storerooms and compounds. Some of these objects were sold or exchanged on the coast. However, many were brought to the UK where they were sold through private auction, donated to museums, or retained by soldiers of the expedition. [. . .]
Maqdala, 1868 (36)	Af1868,1001.29	1868	Looted by Richard Holmes at Maqdala (13 April 1868). After the military assault on Maqdala, official orders were given for all loot to be handed over to the Army Prize Committee. However, Holmes was given permission by Robert Napier (Commander in Chief of the Abyssinian Campaign) to retain some objects (described as 'prizes'). [. . .]
Anglo-Asante wars, 1873–74 / 1895–96 (28)	Af1900,0427.2.a-c	1900	Taken from the royal palace in Kumasi by British troops during the fourth Anglo-Ashanti (Anglo-Asante) War, 1895-6. Formed part of the official government loot. Associated with the Asantehene Prempeh I and recorded as an item of his regalia (with items numbered Af1900,0427.1, 3-10, 12-22). [. . .]
Yok Obolo shrine, 1904 (11)	Af1905,0413.62	1905	In September 1904 British forces led by Alfred Ashmall Whitehouse destroyed the Yok Obolo shrine and confiscated objects from it. The shrine belonged to the Obolo peoples. See Collection File Af1905,0413, previously EthDoc 209, for report by Whitehouse, Colonial Government of Southern Nigeria, 6.9.1904.

▷ Back

Recorded Provenance, City and Grid

	Number of Artifacts Acquired This Year				
	(1) Full sample	(2) Military source	(3) War-related	(4) Explicit loot	(5) None of (2)–(4)
Panel A: City					
Previous Year Had Any British Battle	2.150***	2.654***	4.237***	4.693***	2.066***
	(0.534)	(0.973)	(0.462)	(1.132)	(0.638)
Number of Observations	198,450	198,450	198,450	198,450	198,450
Mean of dep. var.	0.704	0.036	0.010	0.002	0.661
SD of dep. var.	18.460	3.139	0.761	0.455	18.154
Panel B: 1° Grid					
Previous Year Had Any British Battle	2.413***	2.719***	4.265***	4.693***	2.431***
	(0.605)	(0.877)	(0.384)	(1.132)	(0.710)
Number of Observations	213,248	213,248	213,248	213,248	213,248
Mean of dep. var.	0.655	0.033	0.009	0.002	0.615
SD of dep. var.	17.778	3.011	0.613	0.439	17.489
City or Grid FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

▷ Back

What the Provenance Labels Cover

Denominator: 2,172 artifacts acquired from battle cities in a battle year or the next three years. Overlapping windows count once.

- Military source: 334 (15.4%).
- War-related notes: 478 (22.0%).
- Explicit taking: 345 (15.9%).
- Military source or war-related: 737 (34%).

- The groups overlap. Explicit taking belongs to war-related notes.
- The source can name a later seller. Israel Russell bought a figure at a sale of palace spoils and sold it to the Museum. Its source names him (Tythacott, 2018b).

These are shares of observed acquisitions in the windows. The provenance regressions estimate changes in counts.

▷ definitions ▷ the Museum's own notes ▷ Back

How the Provenance Sets Are Built

- Only the acquisition and excavation notes are read. Curators' comments mostly describe the scene a print or medal depicts.
- **Military source**: a named party in the acquisition field holds a British military rank.
- **War-related** (1,886 in the sample): an event construction ("siege of", "punitive expedition"), a campaign name, a regiment or a ship, or a conduit institution such as the Foreign Office. Bare city names and generic "expedition" never qualify.
- **Explicit loot** (385): taking vocabulary (loot, plunder, booty, prize, captured) or directional syntax ("taken from") together with a war word. Every distinct note text was read by hand. Benin 1897 accounts for 311, Magdala 36, the Asante wars 28, the Yok Obolo shrine 11.
- The flag records the Museum's own description of a taking. Counting only the artifacts the Museum marks as taken would understate the battle-related inflow four- to seven-fold.

▷ data ▷ Back

Turning Spoils into Cash: Prize Sales

- 33 prize funds for named campaigns. The 20 with a stated total add up to about £3.4 million (US$700 million at 2025 prices).
- Nearly all came from selling captured jewels, specie, and plate.
- Lord Combermere took £60,000 from Bharatpur (India) in 1826 (Biddulph, 1933).
- Prize sales were fast.
  - The Museum's Magdala purchases entered the collection that year.
  - The Kumasi (Ghana) regalia reached Garrard's within three months, and 16 pieces reached the Museum that year (Watt, 2023).

▷ prize funds ▷ Back

Prize Funds, 1816–1913 (1/2)

Fund (campaign, years)	Record	Fund total (£, nominal)	Property sold	Commander's share (£, nominal)	In sample
Deccan booty (Maratha War, 1817–18)	notices 1826–50, shares	850,000[a]	cash and treasure	eighth, about 100,000	1
Ras al-Khaimah (Persian Gulf, 1819–20)	booty; notices 1828	—	booty and treasure	captain's class 458	0
Ava (First Burma War, 1824–26)	notices 1840–45	—	unknown	—	1
Bhurtpore (Siege of Bharatpur, 1826)	Warrant 1826; notices 1830	480,000[a]	treasure in the fort	Combermere 60,000	1
Coorg (Coorg campaign, 1834)	notice 1837, Chelsea	—	unknown	—	0
Ghuznee (Capture of Ghazni, 1839)	notices 1849–50 (LG 21038)	—	unknown	—	0
Kelat and Nepanee (Kalat, 1839–41)	notices 1848, Chelsea	—	unknown	—	1
Syria and Acre (Egyptian War, 1840)	gratuity; notices 1844	—	none, a gratuity	Stopford 2,412	0
Jellalabad (Defence of Jalalabad, 1842)	Warrant 1842, India	—	unknown	—	1
China batta (First Opium War, 1841–42)	notices 1844–45, EIC	—	none, a gratuity	—	6
Scinde (Conquest of Sind, 1843)	notices 1849–52 (LG 21125)	540,000[b]	the Amirs' treasure	Napier 68,000	1
Sutlej (First Sikh War, 1845–46)	Warrant 1845, India	—	unknown	—	2
Punjab (Second Sikh War, 1848–49)	Warrant 1851, Calcutta Gazette	130,992[d]	jewels, other property, ordnance	—	2
Pegu (Second Burma War, 1852–53)	Chelsea rolls, 1861–64	—	unknown	—	2
Kertch (Crimean War, 1855)	grant; notices 1864	85,925[c]	stores, valued	each flag officer 2,969	1
Persia (Anglo-Persian War, 1856–57)	India Office prize rolls	—	unknown	—	1
Canton (Second Opium War, 1857–58)	Treasury award 1860	50,544[c]	value awarded	—	3
Delhi (Indian Mutiny, 1857)	Warrant 1861, LG 22478	346,121	jewels, coin, shawls, arms	17,306, four commanders	1

▷ next ▷ data ▷ Back

Prize Funds, 1816–1913 (2/2)

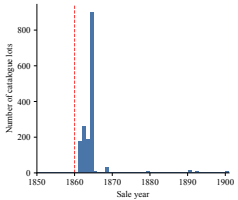
Delhi (Indian Mutiny, 1857)	Warrant 1861, LG 22478	348,121	jewels, coin, shawls, arms	17,366, four commanders	1
Lucknow (Indian Mutiny, 1857–58)	Warrant 1861, LG 22478	148,265	gold, jewels, shawls, dresses	Clyde 7,413	3
Dhar (Indian Mutiny, 1857)	Warrant 1862, LG 22675	77,101	coin, cannon, elephants	—	2
Hulgullee (Indian Mutiny, 1857)	Warrant 1862, LG 22675	490	cattle	—	0
Sarun (Indian Mutiny, 1858)	Warrant 1862, LG 22675	4,960	cash, grain, elephants, cattle	—	0
Shorapore (Indian Mutiny, 1858)	Warrant 1862, LG 22675	6,831	arms, animals, jewels	—	0
Nurgoond (Indian Mutiny, 1858)	Warrant 1862, LG 22675	1,003	livestock and grain	—	0
Khandeish (Indian Mutiny, 1858)	Warrant 1862, LG 22675	11,709	silver, coin, ornaments	—	0
Banda and Kirwee (Indian Mutiny, 1858)	Warrant 1867, LG 23216	555,000	specie, jewels, diamonds	Clyde 22,200; Whitlock 5,550	1
Jhansi (Indian Mutiny, 1858)	Warrant 1867, LG 23216	52,212	property sold	twentieth to Clyde and Rose	3
Kotah (Indian Mutiny, 1858)	Warrant 1867, LG 23216	12,498	property sold	—	1
Pekin (Second Opium War, 1860)	field auction, not gazetted	26,000[a]	palace loot and treasure	generals waived their shares	1
Abyssinia (Magdala expedition, 1868)	gazetted 1872	2,000	loot; crown and chalice	—	1
Kumasi (Third Ashanti War, 1873–74)	prize sales, not gazetted	11,000[c]	gold regalia	—	0
Mandalay (Third Burma War, 1885)	spot auction, not gazetted	—	palace contents	—	1
Peking (Boxer Rebellion, 1900)	auctions; LG 27492 (1902)	—	loot auctions	—	2
Total, 20 funds with a figure, nominal		3,392,651			
in 1900 pounds		3,063,783			
at 2025 prices		529,035,370			

▷ data ▷ Back

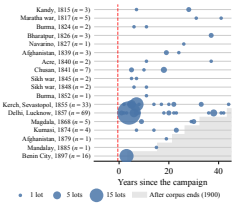
Turning Spoils into Cash: The Salerooms

- 14.5 million lots in 27,751 sale catalogues, 1750–1900: 1,803 lots sold as taken in war, from 19 campaigns.
- Yuanmingyuan (Old Summer Palace, Beijing, 1860): 1,643 lots over decades, against 13 Museum artifacts from Beijing within four years.

Yuanmingyuan lots by sale year



Other campaigns, years since the campaign



▷ catalogue pages ▷ salerooms and Museum ▷ catalogue coverage and prices ▷ Back

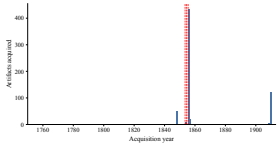
Campaign Lots in the Salerooms and the Museum

Campaign	Saleroom lots		British Museum objects			
	Sold as campaign objects	Military-titled consignor	Acquired within four years	Of which explicit-loot notes	Acquired 21 or more years later	Median lag (years)
Kerch and Sevastopol, 1855	33	1	462	0	131	2
Delhi, 1857	11	2	0	0	5	23
Lucknow, 1857–58	50	45	0	0	6	33.5
Yuanmingyuan (Beijing), 1860	1,643	491	13	0	843	34
Magdala, 1868	5	1	49	28	3	0
Kumasi, 1874	4	0	122	3	339	22
Benin City, 1897	16	16	302	292	—	1

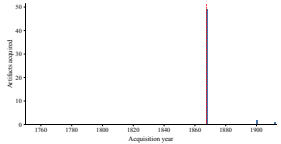
▷ Back

Four Campaigns and Their Acquisition Channels

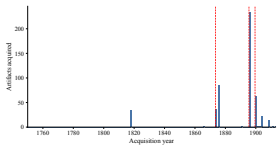
Sevastopol and Kerch: officer donations



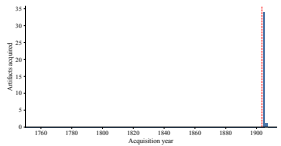
Magdala: army sale



Kumasi: auction and government transfer

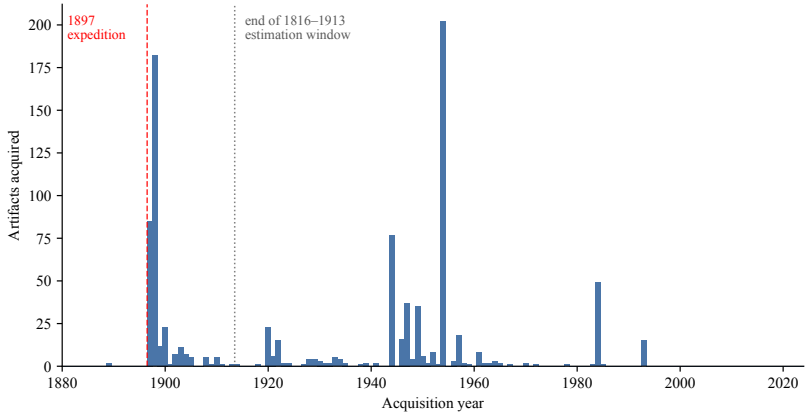


Gyantse: expedition acquisitions



▷ channels and timing ▷ Back

Benin City to 2024



▷ Back

Inputs to the Back-of-the-Envelope Calculation

Years after the battle (h)	Coefficient ($\hat{\beta}_h$)	City-years (N_h)	Mean acquisitions ($\bar{y}_h$)	$N_h \bar{y}_h (1 - e^{-\hat{\beta}_h})$
0 (battle year)	0.62	316	0.97	141
1	2.10	270	5.55	1,315
2	0.83	263	0.89	133
3	−0.11	259	0.51	−16
Total		1,108		1,573

- Without the battle, the Museum would have acquired $\bar{y}_h e^{-\hat{\beta}_h}$ per city-year h years after it. Summing $N_h \bar{y}_h (1 - e^{-\hat{\beta}_h})$ over $h = 0, \ldots, 3$ gives about 1,570.
- Where a city had a second battle within four years, only the most recent battle's multiplier is applied, which understates the total.
- The 1,570 are 4.2% of the 37,276 artifacts the Museum has acquired from battle cities in any year.

▷ Back

Valuing the Artifacts at Historical Auction Prices

- Training data: realized prices from the Getty Provenance Index and priced lots from the sale catalogues, 261,122 lots, converted to pounds at the sale year and to 1900 purchasing power.
- A multilingual BERT model predicts log price from the printed description and the sale context. Museum records enter through description fields only, with prices and acquisition details withheld.
- The 2,172 artifacts acquired from battle cities within four years of a battle score £21,609 together. The 1,570 attributed artifacts at equal value: £15,600, 20% of the valuation of all battle-city acquisitions in the period.
- Validation on 488 Museum artifacts with a matched sale: median multiplicative error 4.96, one in five within a factor of two, and a held-out R^2 of 0.33. The model serves shares and totals, never single records.
- Unvalued: officers' cash receipts, objects outside the Museum, and the loss to the societies of origin.

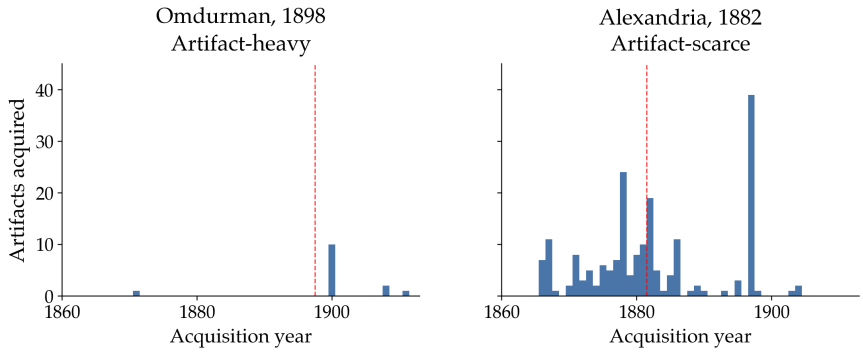
▷ Back

How Long After a Battle Did Artifacts Arrive?

- For the 35,840 artifacts the Museum acquired from a battle city in or after the year of its first battle, the median lag since the most recent battle is 89 years. 6.1% arrive within the four-year window.
- Artifacts whose notes record a taking arrive with a median lag of one year, 77% within the window. War-related notes: two years, 64%. Military-titled sources: 25 years, 21%.
- Dividing the attributed 1,583 by these shares brackets the eventual inflow at 2,050 to 7,722 artifacts, under the assumption that the attributable flow shares the arrival profile of the marked artifacts.
- The long-run estimates measure the same accumulation: a battle city sent 2.8 artifacts in the year after its first battle and 31 over the five decades after the battle decade.

▷ Back

Classifying Battles by Museum Inflows



Annual acquisitions. Dashed lines mark the battles. The strict classification compares the three years before and after each battle, excluding the battle year.

▷ Back

Winsorized and Trimmed Estates

	Probate Wealth at Death (£ thousand, 1900 prices)			
	Winsorized at 99th pct.		Trimmed at 99th pct.	
	(1)	(2)	(3)	(4)
Panel A: 3-Year Window				
Number of artifact-heavy (strict 2× pre) battles	12.276*	14.294*	10.534*	12.954**
	(6.620)	(7.367)	(5.852)	(6.279)
Number of artifact-heavy (any post-inflow) battles	6.879***	8.399***	5.810***	7.091***
	(2.241)	(2.713)	(2.024)	(2.333)
Number of artifact-heavy (above median, inflow > 0) battles	13.645**	15.378**	12.077**	14.424**
	(6.622)	(7.282)	(5.903)	(6.225)
Panel B: 20-Year Window				
Number of artifact-heavy (strict 2× pre) battles	10.651***	11.634***	8.595***	9.229***
	(3.317)	(3.636)	(2.984)	(2.896)
Number of artifact-heavy (any post-inflow) battles	6.429***	6.983***	5.673***	6.306***
	(1.889)	(2.000)	(1.868)	(1.898)
Number of artifact-heavy (above median, inflow > 0) battles	9.715***	10.632***	8.143***	8.829***
	(2.878)	(3.128)	(2.628)	(2.519)
Background and Rank Controls	No	Yes	No	Yes
Number of Observations	218	218	215	215
Mean of dep. var. (£k, 1900)	19.83	19.83	17.70	17.70
SD of dep. var. (£k, 1900)	31.83	31.83	26.37	26.37

▷ Back

Logs and Poisson

	Log Probate Wealth at Death (1900 prices)		Probate Wealth at Death (£ thousand, PPML)	
	(1)	(2)	(3)	(4)
Panel A: 3-Year Window				
Number of artifact-heavy (strict 2× pre) battles	0.323	0.375	0.417**	0.558**
	(0.283)	(0.284)	(0.168)	(0.218)
Number of artifact-heavy (any post-inflow) battles	0.203*	0.232	0.215***	0.401***
	(0.120)	(0.146)	(0.048)	(0.098)
Number of artifact-heavy (above median, inflow > 0) battles	0.398	0.421	0.453***	0.578***
	(0.281)	(0.277)	(0.163)	(0.210)
Panel B: 20-Year Window				
Number of artifact-heavy (strict 2× pre) battles	0.295**	0.339**	0.352***	0.393***
	(0.141)	(0.148)	(0.068)	(0.112)
Number of artifact-heavy (any post-inflow) battles	0.204**	0.240**	0.218***	0.265***
	(0.084)	(0.094)	(0.044)	(0.060)
Number of artifact-heavy (above median, inflow > 0) battles	0.244*	0.283**	0.332***	0.382***
	(0.133)	(0.137)	(0.061)	(0.105)
Background and Rank Controls	No	Yes	No	Yes
Number of Observations	217	217	218	218
Mean of dep. var.	1.85	1.85	20.42	20.42
SD of dep. var.	1.68	1.68	35.37	35.37

▷ Back

Accounting Windows from 2 to 50 Years

	Probate Wealth at Death (£ thousand, 1900 prices)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Accounting window (years)	2	3	4	5	20	30	40	50
Number of artifact-heavy battles:								
strict 2× pre	16.165*	14.380*	11.358*	9.594*	13.390***	10.832***	10.997***	7.806***
	(8.643)	(7.704)	(6.037)	(5.019)	(4.811)	(3.740)	(3.399)	(2.430)
any post-inflow	6.599**	9.238***	10.438**	10.792***	7.432***	7.423***	6.860***	6.160***
	(3.114)	(3.084)	(4.172)	(3.827)	(2.187)	(1.892)	(1.704)	(1.869)
above median, inflow > 0	17.263**	15.381**	13.840**	9.989**	12.068***	10.254***	10.852***	10.350***
	(8.481)	(7.612)	(5.699)	(4.903)	(4.052)	(3.336)	(3.250)	(3.013)
Artifact-heavy battles:								
strict 2× pre	33	37	40	44	80	103	127	147
any post-inflow	70	81	93	105	196	177	221	217
above median, inflow > 0	35	39	45	52	98	122	135	137
British battles	474	474	474	474	474	474	474	474
Background and Rank Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of Observations	218	218	218	218	218	218	218	218
Mean of dep. var. (£k, 1900)	20.42	20.42	20.42	20.42	20.42	20.42	20.42	20.42
SD of dep. var. (£k, 1900)	35.37	35.37	35.37	35.37	35.37	35.37	35.37	35.37

▷ Back

Extensive and Intensive Margins

	Probate Wealth at Death (£ thousand, 1900 prices)	
	(1)	(2)
Panel A: 3-Year Window		
Number of artifact-heavy (strict 2× pre) battles	12.252*	14.380*
	(6.848)	(7.704)
Any artifact-heavy (strict 2× pre) battle	18.817*	20.069*
	(9.572)	(10.367)
asinh(post-battle BM inflow)	4.849**	5.016**
	(1.947)	(2.191)
asinh(net post-battle BM inflow)	4.195**	4.283**
	(1.753)	(1.931)
Panel B: 20-Year Window		
Number of artifact-heavy (strict 2× pre) battles	12.103***	13.390***
	(4.155)	(4.811)
Any artifact-heavy (strict 2× pre) battle	15.891**	16.957**
	(7.754)	(8.365)
asinh(post-battle BM inflow)	2.825**	3.021**
	(1.255)	(1.334)
asinh(net post-battle BM inflow)	1.927*	2.104**
	(1.009)	(0.996)
Background and Rank Controls	No	Yes
Number of Observations	218	218
Mean of dep. var. (£k, 1900)	20.42	20.42
SD of dep. var. (£k, 1900)	35.37	35.37

▷ Back

Lee Bounds on the Probate Linkage

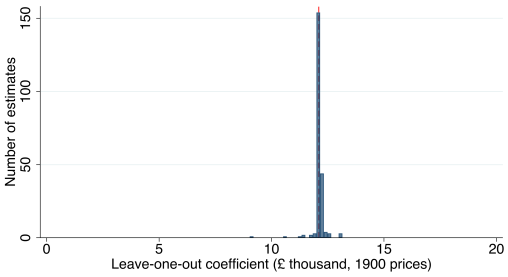
	Probated Wealth (£, 1900 prices)	
Artifact-heavy definition:	(1) Strict (2× pre), 3-year window	(2) Strict (2× pre), 20-year window
Lower bound	19,882*	16,835**
	(10,358)	(8,330)
Baseline	19,981*	16,906**
	(10,366)	(8,362)
Upper bound	22,805**	18,323**
	(10,125)	(8,226)
Match rate, artifact-heavy commanders	0.879	0.883
Match rate, other commanders	0.891	0.891
Trimming share	0.014	0.009
Number of Observations	217	217

- Of the 27 unmatched commanders, 13 were colonial officers who lived and died overseas, and 14 left no grant under their name after extensive search.

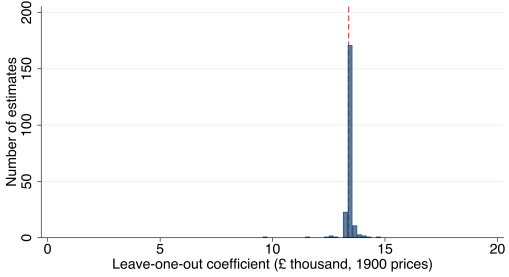
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Dropping One Commander at a Time

No controls



Background and rank controls



▷ Back

Commanders and Other Officers

Leading a battle paid on its own, and the artifact-heavy premium came on top.

Dep. Var.:	Probate Wealth at Death (£ thousand, 1900 prices)		
	(1)	(2)	(3)
Any battle commander vs. Officers in probate	17.238***	14.773***	15.233***
	(3.413)	(3.372)	(3.391)
Artifact-heavy battle commanders vs. Officers in probate	33.693***	31.767***	31.871***
	(12.332)	(12.180)	(12.218)
Other battle commanders vs. Officers in probate	12.939***	10.318***	10.859***
	(2.706)	(2.689)	(2.702)
Rank-Group-at-Death FE	No	Yes	Yes
Death-Decade FE	No	No	Yes
Number of Observations	13,696	13,696	13,696
Mean of dep. var. (£k, 1900)	7.93	7.93	7.93
SD of dep. var. (£k, 1900)	26.17	26.17	26.17

13,556 officers in the 1858–1906 calendars against 140 commanders. With rank at death and death decade: +£31,900 and +£10,900.

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Prize Receipts of Commanders (1/2)

Commander	Fund	Source	Receipt (£, 1900)	Probate (£, 1900)	Prize to probate (%)
Tier 1: Sums stated for the commander					
Combermere	Bhurtpore	Biddulph (1933, p. 230)	55,758	16,356	340.9
Colin Campbell	Banda and Kirwee, Lucknow, Jhansi, Kotah	*London Gazette* 22478, 23216	29,332	102,222	28.7
George Whitlock	Banda and Kirwee	*London Gazette* 23216, 23341–23429	10,350	8,280	125.0
Harry Smith	Sutlej	Military Letter 24 of 1846	2,356	5,935	39.7
Edmund Lyons	Kertch and Yenikale	Lords, 7 Mar 1864, *London Gazette* 22814	1,476	45,495	3.2
Hugh Rose	Jhansi	*London Gazette* 23216, 23361	753	70,758	1.1
Thomas Willshire	Kelat	*Allen's Indian Mail* 6 (1848), p. 631	605	24,731	2.4
Willoughby Cotton	Ghuznee	IOR L/MIL/5/259	367	49,462	0.7
Charles Fitzgerald	Deccan	*London Gazette* 18447, 18962, 20739	216	46,000	0.5
Francis Rowcroft	Sarun	*London Gazette* 22675	203	1,423	14.3
Charles Stuart	Jhansi	*London Gazette* 23216	186	14,154	1.3
Henry Roberts	Kotah	*London Gazette* 23216	114	2,968	3.8
Michael Smith	Jhansi	*London Gazette* 23216	62	15,387	0.4
John Haughton	Ghuznee	IOR L/MIL/5/259	15	3,614	0.4

▷ next ▷ Back

Prize Receipts of Commanders (2/2)

Tier 2: Sums imputed from rank at the battle					
Archdale Wilson	Delhi	*London Gazette* 22478	11,292	3,680	306.8
Michael Seymour	Canton 1857	*London Gazette* 22479	1,397	30,635	4.6
Edward Greathed	Delhi	*London Gazette* 22478, 22594	289	17,673	1.6
James Outram	Lucknow	*London Gazette* 22478, 22597	264	30,667	0.9
Robert Napier	Abyssinia	*London Gazette* 23879	230	9,398	2.4
Thomas Seaton	Delhi	*London Gazette* 22594, 22649	145	1,408	10.3
William Mansfield	Lucknow	*London Gazette* 22478	128	56,327	0.2
Adrian Hope	Lucknow	*London Gazette* 22478	86	303	28.3
Alfred Horsford	Lucknow	*London Gazette* 22478	86	11,310	0.8
Frederick Evelegh	Lucknow	*London Gazette* 22478	86	71,334	0.1
Robert Walpole	Lucknow	*London Gazette* 22478	86	32,857	0.3
Thomas Franks	Lucknow	*London Gazette* 22478	86	3,957	2.2
William Peel	Lucknow	*London Gazette* 22478, 23017	86	35,385	0.2
Henry Havelock	Lucknow	*London Gazette* 22478, 22597	30	1,380	2.1
Frederick Roberts	Kotah	*London Gazette* 23216, 23308	8	72,571	0.0

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Prize Receipts, Sources and Gaps

- Gazette ladders price a share where a warrant prints a distribution and the commander led a battle it names. That leaves 64 commanders in a fund campaign without a priced share.
- Search: Hansard, the *Calcutta Gazette*, the India Office prize rolls, *Allen's Indian Mail*, regimental histories, and biographies. Nineteen are priced, nine are stated only by class, and nine sit in archives not consulted.
- Tier 1: fourteen sums a source states for the man himself. Tier 2: fifteen computed from a per-rank ladder and the rank at the battle. 25 fought where no fund existed, zeros in the record rather than gaps.
- Over the 29, documented prize equals 15% of combined probate and 25% in tier 1. Four commanders-in-chief hold 92% of the total, two-thirds drew less than £300, and the mean ratio to probate is 0.32 against a median of 0.02.
- Lower bounds: booty settled on the spot never reached the Gazette, and later installments gazetted without a rate are not counted.

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How Commanders Got Rich: Sales and Careers

- Kept objects: Captain Charles Gordon valued a Yuanmingyuan screen at no less than £1,000, and Colonel Luard sold twenty Yuanmingyuan enamels in London in 1861.
- Careers: an artifact-heavy battle raised the chance of a peerage by 12.9 pp (mean 7.3%) and of high office by 9.3 pp (mean 28%).

	Peerage at death		High office	
	(1)	(2)	(3)	(4)
Number of artifact-heavy battles led	0.135**	0.129***	0.005	0.093**
	(0.062)	(0.029)	(0.078)	(0.039)
Window	3-year	20-year	3-year	20-year
Background and rank controls	Yes	Yes	Yes	Yes
Number of Observations	218	218	218	218
Mean of dep. var.	0.073	0.073	0.284	0.284

- Time: commanders who died within three years of their last battle left smaller estates. ▷ go

▷ Back

Death Timing and Wealth

	Probate Wealth at Death (£ thousand, 1900 prices)	
	(1)	(2)
Died within 3 years of last battle	-9.441*	-8.466*
	(5.382)	(5.066)
Died at battle	-10.315*	-8.253
	(6.121)	(6.327)
Died at a lost battle	-15.087***	-13.030*
	(5.529)	(6.667)
Background and Rank Controls	No	Yes
Number of Observations	218	218
Mean of dep. var. (£k, 1900)	20.42	20.42
SD of dep. var. (£k, 1900)	35.37	35.37

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Commanders' Wealth: Identification

Identifying assumption

Conditional on the observed characteristics, assignment to artifact-heavy rather than artifact-scarce campaigns was unrelated to a commander's latent wealth prospects.

- **Ability.** Abler officers might lead important battles. Artifact-heavy tracks what a place held, and need not track military importance.
- **Choice.** Officers could seek profitable campaigns, or superiors could favor them. In 31% of the 406 commander–battle pairs, the commander already held the standing command in or next to the theatre.
- In war the Secretary of State chose the commander-in-chief, and the Commander-in-Chief recommended the others (Biddulph, 1904; Beckett, 2018).

▷ assignment evidence and cases ▷ Back

Did Officers Choose Their Campaigns?

- Importance is not artifact-heaviness. The 1837 battles in Lower Canada generated no inflow, and Colborne, who rose to field marshal and a peerage, left £5,100 against £49,000 for the nineteen commanders who died field marshals.
- Commanders often fought because they were posted nearest the enemy. In 31% of the 406 commander-battle pairs the commander already held the standing command in or next to the theatre, read from Gazette appointment notices, campaign histories, and biographical dictionaries.
- Admiral Rawson led the Benin expedition because the killing of the acting consul-general's party fell inside the Cape station, which he had held since 1895.
- No officer could appoint himself. In war the Secretary of State chose the commander-in-chief, and the Commander-in-Chief recommended the others (Biddulph, 1904; Beckett, 2018).
- Inside the incident-triggered subset, the artifact-heavy estimate is also positive and not distinguishable from the rest ($p = 0.28$).

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Why the Mutiny?

- Regimental campaign histories overlap heavily. Most regiments cycled through several wars within a decade.
- A design pooling every campaign would compare newly treated regiments with regiments still responding to their own recent campaigns.
- The Mutiny's prize was among the largest of the century, and its regiments can be separated from the Crimean War regiments.
- A second campaign, the First Afghan War of 1839–42 with its gazetted Ghuznee and Kelat prize, gives a similar rise. ▷ go

▷ Back

The Treated and Control Regiments

- Purchasable line and cavalry regiments with sustained Gazette coverage, classified from the battle rosters, the Gazette dispatches of 1848–1870, and regimental histories. Artillery, engineers, marines, and Company units are out.
- **Treated (25)**: 2nd and 6th Dragoon Guards, 7th and 14th Hussars, 9th and 12th Lancers, and the 5th, 8th, 10th, 13th, 32nd, 37th, 52nd, 53rd, 60th, 61st, 64th, 72nd, 75th, 78th, 80th, 82nd, 83rd, 84th, and 86th Foot. 19 are named in the campaign dispatches.
- **Control (31)**: Life Guards, Royal Horse Guards, 7th Dragoon Guards, 10th and 15th Hussars, 16th Lancers, the three West India Regiments, and the 11th, 15th, 16th, 22nd, 25th, 27th, 36th, 39th, 48th, 54th, 56th, 58th, 66th, 69th, 76th, 81st, 85th, 87th, 89th, 94th, 96th, and 98th Foot. Exclusion from control requires positive evidence of fighting.
- Out of both: sixteen Crimean-veteran Mutiny regiments, the 35th and 92nd Foot (no Mutiny honor), and three regiments with dispatch namings elsewhere.
- Battle honors agree with the assignment for 54 of 56 regiments. ▷ go

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Battle Honors Benchmark

Panel A. Treated regiments (substantial Mutiny service), $N = 25$

Regiment	Mutiny honour	Agrees	Regiment	Mutiny honour	Agrees
2nd Dragoon Guards	Lucknow	Yes	53rd Foot	Lucknow	Yes
6th Dragoon Guards	Delhi 1857	Yes	60th Foot	Delhi	Yes
7th Hussars	Lucknow	Yes	61st Foot	Delhi 1857	Yes
14th Hussars	Central India	Yes	64th Foot	Lucknow	Yes
9th Lancers	Delhi 1857, Lucknow	Yes	72nd Foot	Central India	Yes
12th Lancers	Central India	Yes	75th Foot	Delhi 1857, Lucknow	Yes
5th Foot	Lucknow	Yes	78th Foot	Lucknow	Yes
8th Foot	Delhi 1857, Lucknow	Yes	80th Foot	Central India	Yes
10th Foot	Lucknow 1858	Yes	82nd Foot	Lucknow	Yes
13th Foot	None	No	83rd Foot	Central India	Yes
32nd Foot	Lucknow	Yes	84th Foot	Lucknow	Yes
37th Foot	None	No	86th Foot	Central India	Yes
52nd Foot	Delhi 1857	Yes			

Panel B. Control regiments (no substantial combat 1848–1870), $N = 31$

Regiment	Mutiny honour	Agrees	Regiment	Mutiny honour	Agrees
Life Guards	None	Yes	39th Foot	None	Yes
Royal Horse Guards	None	Yes	48th Foot	None	Yes
7th Dragoon Guards	None	Yes	54th Foot	None	Yes
10th Hussars	None	Yes	56th Foot	None	Yes
15th Hussars	None	Yes	58th Foot	None	Yes
16th Lancers	None	Yes	66th Foot	None	Yes
1st West India Regiment	None	Yes	69th Foot	None	Yes
2nd West India Regiment	None	Yes	76th Foot	None	Yes
3rd West India Regiment	None	Yes	81st Foot	None	Yes
11th Foot	None	Yes	85th Foot	None	Yes
15th Foot	None	Yes	87th Foot	None	Yes
16th Foot	None	Yes	89th Foot	None	Yes
22nd Foot	None	Yes	94th Foot	None	Yes
25th Foot	None	Yes	96th Foot	None	Yes
27th Foot	None	Yes	98th Foot	None	Yes
36th Foot	None	Yes			

Treated regiments holding a Mutiny honour: 23 of 25. Control regiments holding none: 31 of 31. Arm and honour agree for 54 of the 56 regiments.

23 of the 25 treated regiments hold a Mutiny honor, and none of the 31 controls does. The 13th and 37th Foot are named in the campaign dispatches but received no honor.

▷ regiments ▷ Back

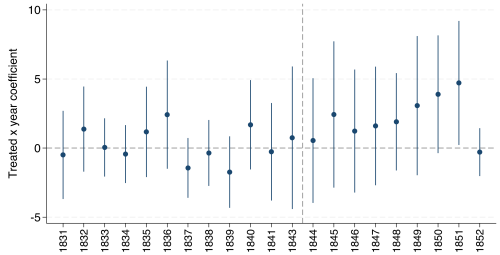
Mutiny Difference-in-Differences

	Gazetted Officer-Career Events per Regiment-Year			
	(1) All purchase events	(2) Promotions by purchase	(3) First commissions purchased	(4) Within-regiment promotions
Panel A: Mutiny-only regiments vs. no-substantial-combat controls (strict)				
Treated × Post (1859–68)	1.850**	1.053**	0.797**	1.104**
	(0.829)	(0.475)	(0.379)	(0.476)
Number of Observations	1,176	1,176	1,176	1,176
Mean of dep. var.	5.913	3.556	2.357	3.465
SD of dep. var.	4.854	3.155	2.063	3.106
Mean of dep. var. (treated, pre-1859)	4.098	2.505	1.593	2.389
Panel B: All Mutiny regiments vs. all other purchasable regiments (broad)				
Treated × Post (1859–68)	1.001*	0.634*	0.367	0.650*
	(0.589)	(0.343)	(0.263)	(0.348)
Number of Observations	2,667	2,667	2,667	2,667
Mean of dep. var.	5.961	3.574	2.387	3.494
SD of dep. var.	4.653	3.065	1.979	3.018
Mean of dep. var. (treated, pre-1859)	4.569	2.763	1.806	2.670
Regiment FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

▷ Back

Robustness and a Second Campaign

- Other comparison groups. ▷ go
- Poisson, asinh, winsorized, normalized, and wild cluster bootstrap. ▷ go
- Drop any one of 56 regiments: 1.59 to 2.14. ▷ go
- First Afghan War (1839–42): a rise of 46% of the pre-campaign rate, against 45% for the Mutiny. ▷ go ▷ design

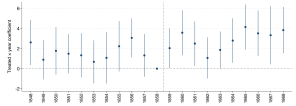


First Afghan War: commission purchases

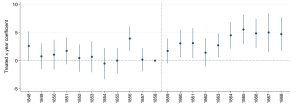
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Mutiny, Alternative Comparison Sets

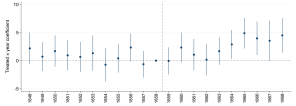
All Mutiny regiments vs. regiments that fought elsewhere



Mutiny-only regiments vs. regiments that fought elsewhere



Mutiny-only regiments vs. regiments untouched by the Crimean and Beijing campaigns



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Mutiny Robustness (1/2)

	All Purchase Events per Regiment-Year		
Panel A: Alternative comparison sets			
	(1) All Mutiny vs. other fighters	(2) Mutiny-only vs. other fighters	(3) Mutiny-only vs. no Crimea/Beijing
Treated × Post (1859–68)	1.371**	2.675***	1.685**
	(0.598)	(0.669)	(0.775)
Number of Observations	2,121	1,848	1,449
Mean of dep. var.	5.873	5.970	5.955
SD of dep. var.	4.598	4.631	4.808
Panel B: Alternative estimators			
	(1) OLS	(2) Poisson	(3) Inverse hyperbolic sine
Treated × Post (1859–68)	1.850**	0.355***	0.403***
	(0.829)	(0.128)	(0.148)
Number of Observations	1,176	1,176	1,176
Mean of dep. var.	5.913	5.913	2.121
SD of dep. var.	4.854	4.854	0.957

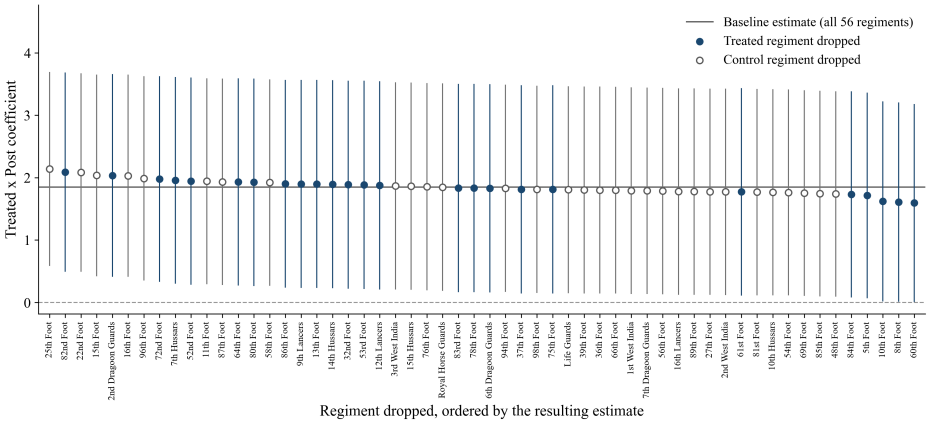
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Mutiny Robustness (2/2)

Panel C: Outcome and inference			
	(1) Winsorized at the 99th percentile	(2) Purchases per 100 pre-campaign officer events	(3) Wild cluster bootstrap
Treated × Post (1859–68)	1.750**	8.416**	1.850**
	(0.800)	(3.974)	–
p-value	0.033	0.039	0.031
Number of Observations	1,176	1,176	1,176
Number of Regiments	56	56	56
Panel D: Sample composition			
	(1) Cavalry excluded	(2) 6th Dragoon Guards and 72nd Foot excluded	(3) 94th Foot and West India regiments excluded
Treated × Post (1859–68)	2.033**	1.961**	1.696*
	(1.004)	(0.856)	(0.884)
Number of Observations	924	1,134	1,092
Number of Regiments	44	54	52
Regiment FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

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Dropping One Regiment at a Time



▷ Back

The First Afghan War

	(1) First Afghan War 1839–1842	(2) Indian Mutiny 1857–1858
Difference-in-differences, all purchase events	1.819*	1.850**
	(1.049)	(0.829)
Treated pre-campaign mean	3.92	4.10
Estimate as percent of the pre-campaign mean	46	45
Window	1831–1852	1848–1868
Treated regiments	5	25
Control regiments	67	31
Number of Observations	1,584	1,176
Mean of dep. var.	4.824	5.913
SD of dep. var.	3.782	4.854
Regiment FE	Yes	Yes
Year FE	Yes	Yes

▷ the design ▷ Back

The First Afghan War Design

- Treated: the five purchasable regiments with honor-bearing service in 1839–42. The 13th Foot led through the Kabul Gate at Ghuznee and held Jalalabad, the 2nd and 17th supplied the storming parties at Ghuznee and Kelat, the 41st held Kandahar, and the 44th was destroyed on the retreat at Gandamak.
- Control: the 67 purchasable regiments that fought nothing between 1831 and 1852.
- Window: eight years before the campaign to ten after its final year, with regiment and year effects, clustered by regiment.
- Prize: the Ghuznee roll gave £379 to a major-general, about £30 to a captain, and five shillings to a private. The Kelat booty was about £10,000.
- Purchases in the treated regiments rise by 1.82 a year over 1843–52, 46% of their pre-campaign rate, with no pre-campaign trend. ▷ table

▷ Back

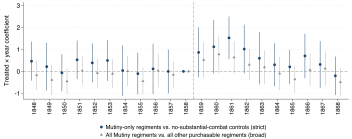
Who Bought, Strict and Broad Comparisons

	Gazetted Officer-Career Events per Regiment-Year					
	Baseline	Attraction		Cash in hand	Release	
	(1) All purchase events	(2) Exchanges into the regiment	(3) First commissions purchased	(4) Within-regiment promotions by purchase	(5) Retirements and resignations	(6) Purchases into retirement vacancies
Panel A: Mutiny-only regiments vs. no-substantial-combat controls (strict)						
Treated × Post (1859–68)	1.850**	0.457***	0.797**	1.104**	0.613*	0.573*
	(0.829)	(0.166)	(0.379)	(0.476)	(0.364)	(0.306)
Number of Observations	1,176	1,176	1,176	1,176	1,176	1,176
Mean of dep. var.	5.91	1.31	2.36	3.47	2.60	2.23
Mean of dep. var. (treated, pre-1859)	4.10	1.03	1.59	2.39	1.84	1.55
Panel B: All Mutiny regiments vs. all other purchasable regiments (broad)						
Treated × Post (1859–68)	1.001*	0.324**	0.367	0.650*	0.371	0.337
	(0.589)	(0.134)	(0.263)	(0.348)	(0.259)	(0.214)
Number of Observations	2,667	2,667	2,667	2,667	2,667	2,667
Mean of dep. var.	5.96	1.17	2.39	3.49	2.63	2.24
Mean of dep. var. (treated, pre-1859)	4.57	0.97	1.81	2.67	2.06	1.72
Regiment FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

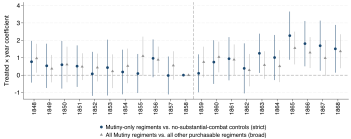
▷ Back

Mutiny Event Studies by Channel

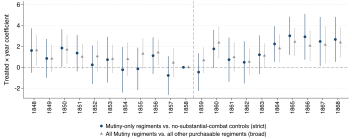
Exchanges into the regiment



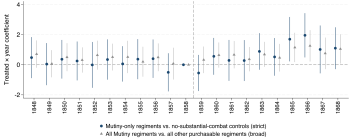
First commissions purchased



Within-regiment promotions by purchase



Retirements and resignations



▷ Back

A Contemporary Account of Lucknow, 1858

- Engels, in the *New-York Daily Tribune*, on the sack of Lucknow: the prize agents "convert loot into a system", and officers and soldiers "went into the town poor and debt-ridden, and came out suddenly enriched" (Engels, 1858).
- Quoting *The Times*: "some of these officers have made, literally, their fortunes", and "above 150 officers have sent in their resignations to Sir Colin Campbell".
- Under purchase, retirement was the moment an officer sold his commission and took his money out. So many applied to leave that the Duke of Cambridge withheld his approval (Bruce, 1980).
- The purchase record tests whether this response is visible at scale. ▷ go

▷ Back

References I

Acemoglu, Daron, Mikhail Golosov, Aleh Tsyvinski, and Pierre Yared, "A Dynamic Theory of Resource Wars," *Quarterly Journal of Economics*, 2012, *127* (1), 283–331.

__ , **Simon Johnson, and James A. Robinson**, "The Colonial Origins of Comparative Development: An Empirical Investigation," *American Economic Review*, 2001, *91* (5), 1369–1401.

Ager, Philipp, Leonardo Bursztyn, Lukas Leucht, and Hans-Joachim Voth, "Killer Incentives: Rivalry, Performance and Risk-Taking among German Fighter Pilots, 1939–45," *Review of Economic Studies*, 2022, *89* (5), 2257–2292.

Aldous, Grahame, "The Law Relating to the Distribution of Prize Money in the Royal Navy and Its Relationship to the Use of Naval Power in War, 1793–1815." PhD dissertation, King's College London 2020.

Allen, Douglas W., "Compatible Incentives and the Purchase of Military Commissions," *Journal of Legal Studies*, 1998, *27* (1), 45–66.

Arkhangelsky, Dmitry, Susan Athey, David A. Hirshberg, Guido W. Imbens, and Stefan Wager, "Synthetic Difference-in-Differences," *American Economic Review*, 2021, *111* (12), 4088–4118.

Ashenfelter, Orley and Kathryn Graddy, "Auctions and the Price of Art," *Journal of Economic Literature*, 2003, *41* (3), 763–786.

Baicker, Katherine and Mireille Jacobson, "Finders Keepers: Forfeiture Laws, Policing Incentives, and Local Budgets," *Journal of Public Economics*, 2007, *91*, 2113–2136.

References II

Barnett, Correlli, *Britain and Her Army, 1509–1970: A Military, Political and Social Survey*, New York: William Morrow, 1970.

Beckett, Ian F. W., *A British Profession of Arms: The Politics of Command in the Late Victorian Army*, Norman: University of Oklahoma Press, 2018.

Bennett, James S., Erin Mutch, Andrew Tollefson, Ed Chalstrey, Majid Benam, Enrico Cioni, Jenny Reddish, Jakob Zsambok, Jill Levine, C. Justin Cook, Pieter Francois, Daniel Hoyer, and Peter Turchin, "Cliopatria – A Geospatial Database of World-Wide Political Entities from 3400BCE to 2024CE," *Scientific Data*, 2025, *12*, 247.

Berman, Nicolas, Mathieu Couttenier, Dominic Rohner, and Mathias Thoenig, "This Mine Is Mine! How Minerals Fuel Conflicts in Africa," *American Economic Review*, 2017, *107* (6), 1564–1610.

Bertrand, Marianne, Robin Burgess, Arunish Chawla, and Guo Xu, "The Glittering Prizes: Career Incentives and Bureaucrat Performance," *Review of Economic Studies*, 2020, *87* (2), 626–655.

Besley, Timothy and Torsten Persson, "The Origins of State Capacity: Property Rights, Taxation, and Politics," *American Economic Review*, 2009, *99* (4), 1218–1244.

__ and __ , *Pillars of Prosperity: The Political Economics of Development Clusters*, Princeton: Princeton University Press, 2011.

References III

__ , **Robin Burgess, Adnan Khan, and Guo Xu**, "Bureaucracy and Development," *Annual Review of Economics*, 2022, *14*, 397–424.

Biddulph, H., "The Era of Army Purchase," *Journal of the Society for Army Historical Research*, 1933, *12* (48), 221–233.

Biddulph, Robert, *Lord Cardwell at the War Office: A History of His Administration, 1868–1874*, London: John Murray, 1904.

Blattman, Christopher and Edward Miguel, "Civil War," *Journal of Economic Literature*, 2010, *48* (1), 3–57.

Bogart, Dan and Latika Chaudhary, "Extractive Institutions? Investor Returns to Indian Railway Companies in the Age of High Imperialism," *Journal of Institutional Economics*, 2019, *15* (5), 751–774.

Borusyak, Kirill, Xavier Jaravel, and Jann Spiess, "Revisiting Event-Study Designs: Robust and Efficient Estimation," *Review of Economic Studies*, 2024, *91* (6), 3253–3285.

Brewer, John, *The Sinews of Power: War, Money and the English State, 1688–1783*, London: Unwin Hyman, 1989.

Bruce, Anthony P. C., *The Purchase System in the British Army, 1660–1871*, London: Royal Historical Society, 1980.

Caygill, Marjorie, *The Story of the British Museum*, 3rd ed., London: British Museum Press, 2002.

Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer, "The Effect of Minimum Wages on Low-Wage Jobs," *Quarterly Journal of Economics*, 2019, *134* (3), 1405–1454.

References IV

Chandler, David and Ian Beckett, *The Oxford Illustrated History of the British Army*, Oxford: Oxford University Press, 1994.

Chen, Jiafeng and Jonathan Roth, "Logs with Zeros? Some Problems and Solutions," *Quarterly Journal of Economics*, 2024, *139* (2), 891–936.

Collier, Paul and Anke Hoeffler, "Greed and Grievance in Civil War," *Oxford Economic Papers*, 2004, *56* (4), 563–595.

Comin, Diego, William Easterly, and Erick Gong, "Was the Wealth of Nations Determined in 1000 BC?," *American Economic Journal: Macroeconomics*, 2010, *2* (3), 65–97.

Coppedge, Michael, John Gerring, Carl Henrik Knutsen, Staffan I. Lindberg, Jan Teorell et al., "V-Dem Country-Year Dataset v16," Varieties of Democracy (V-Dem) Project, 2026. DOI: 10.23696/vdemds26.

Dal Bó, Ernesto and Pedro Dal Bó, "Workers, Warriors, and Criminals: Social Conflict in General Equilibrium," *Journal of the European Economic Association*, 2011, *9* (4), 646–677.

__ , Frederico Finan, and Martín A. Rossi, "Strengthening State Capabilities: The Role of Financial Incentives in the Call to Public Service," *Quarterly Journal of Economics*, 2013, *128* (3), 1169–1218.

__ , Karolina Hutková, Lukas Leucht, and Noam Yuchtman, "Dissecting the Sinews of Power: International Trade and the Rise of Britain's Fiscal-Military State, 1689–1823," *Journal of Economic History*, 2025, *85* (2), 336–369.

References V

Davis, Lance E. and Robert A. Huttenback, *Mammon and the Pursuit of Empire: The Political Economy of British Imperialism, 1860–1912*, Cambridge: Cambridge University Press, 1986.

de Chaisemartin, Clément and Xavier D'Haultfœuille, "Difference-in-Differences Estimators of Intertemporal Treatment Effects," *Review of Economics and Statistics*, 2026, *108* (4), 863–880.

Dell, Melissa, "The Persistent Effects of Peru's Mining *Mita*," *Econometrica*, 2010, *78* (6), 1863–1903.

Dincecco, Mark and Mauricio Prado, "Warfare, Fiscal Capacity, and Performance," *Journal of Economic Growth*, 2012, *17* (3), 171–203.

Dube, Arindrajit, Daniele Girardi, Òscar Jordà, and Alan M. Taylor, "A Local Projections Approach to Difference-in-Differences," *Journal of Applied Econometrics*, 2025, *40* (7), 741–758.

Dube, Oeindrila and Juan F. Vargas, "Commodity Price Shocks and Civil Conflict: Evidence from Colombia," *Review of Economic Studies*, 2013, *80* (4), 1384–1421.

Eggers, Andrew C. and Jens Hainmueller, "MPs for Sale? Returns to Office in Postwar British Politics," *American Political Science Review*, 2009, *103* (4), 513–533.

Engels, Friedrich, "The British Army in India," *New-York Daily Tribune*, 26 June 1858. Reprinted in *Marx & Engels Collected Works*, vol. 15, pp. 556–559.

Fenske, James and Hans-Joachim Voth, "Conflict in Economic History," in Massimo Morelli, Debraj Ray, and Oeindrila Dube, eds., *Handbook of the Economics of Conflict*, Vol. 1, Amsterdam: Elsevier, 2024, pp. 105–163.

References VI

Finan, Frederico, Benjamin A. Olken, and Rohini Pande, "The Personnel Economics of the Developing State," in Abhijit Vinayak Banerjee and Esther Duflo, eds., *Handbook of Economic Field Experiments*, Vol. 2, Amsterdam: North-Holland, 2017, pp. 467–514.

Fisman, Raymond and Shang-Jin Wei, "The Smuggling of Art, and the Art of Smuggling: Uncovering the Illicit Trade in Cultural Property and Antiques," *American Economic Journal: Applied Economics*, 2009, *1* (3), 82–96.

Gallagher, John and Ronald Robinson, "The Imperialism of Free Trade," *Economic History Review*, 1953, *6* (1), 1–15.

Gibbons, Robert, "Incentives Between Firms (and Within)," *Management Science*, 2005, *51* (1), 2–17.

Gourieroux, Christian, Alain Monfort, and Alain Trognon, "Pseudo Maximum Likelihood Methods: Applications to Poisson Models," *Econometrica*, 1984, *52* (3), 701–720.

Guardado, Jenny, "Office-Selling, Corruption, and Long-Term Development in Peru," *American Political Science Review*, 2018, *112* (4), 971–995.

Hansard, "Regiments in India—Question," House of Commons, 3rd ser., vol. 156, cols. 2146–2147 2 March 1860. `https://api.parliament.uk/historic-hansard/commons/1860/mar/02/question-2`.

__ , "Army—Regiments in India—Question," House of Commons, 3rd ser., vol. 182, col. 1509 17 April 1866. `https://hansard.parliament.uk/Commons/1866-04-17/debates/1c3bed97-c712-4d66-bc89-162fb8d93adf/CommonsChamber`.

References VII

Harries-Jenkins, Gwyn, *The Army in Victorian Society*, London: Routledge & Kegan Paul, 1977.

Heblich, Stephan, Stephen J. Redding, and Hans-Joachim Voth, "Slavery and the British Industrial Revolution," Working Paper 30451, National Bureau of Economic Research 2022.

Hicks, Dan, *The Brutish Museums: The Benin Bronzes, Colonial Violence and Cultural Restitution*, London: Pluto Press, 2020.

Hillmann, Henning and Christina Gathmann, "Overseas Trade and the Decline of Privateering," *Journal of Economic History*, 2011, *71* (3), 730–761.

Hobson, J. A., *Imperialism: A Study*, London: James Nisbet, 1902.

Hoffman, Philip T., *Why Did Europe Conquer the World?*, Princeton: Princeton University Press, 2015.

Holmström, Bengt and Paul Milgrom, "Multitask Principal-Agent Analyses: Incentive Contracts, Asset Ownership, and Job Design," *Journal of Law, Economics, and Organization*, 1991, *7* (Special Issue), 24–52.

Huillery, Elise, "The Black Man's Burden: The Cost of Colonization of French West Africa," *Journal of Economic History*, 2014, *74* (1), 1–38.

Hutton, Edward T. H., ed., *A Brief History of the King's Royal Rifle Corps*, Winchester: Warren and Son, 1912.

Jasanoff, Maya, *Edge of Empire: Lives, Culture, and Conquest in the East, 1750–1850*, New York: Knopf, 2005.

Johnson, Noel D. and Mark Koyama, "Tax Farming and the Origins of State Capacity in England and France," *Explorations in Economic History*, 2014, *51*, 1–20.

References VIII

Joseph, John and Metin Sengul, "Organization Design: Current Insights and Future Research Directions," *Journal of Management*, 2025, *51* (1), 249–308.

Khan, Adnan Q., Asim I. Khwaja, and Benjamin A. Olken, "Tax Farming Redux: Experimental Evidence on Performance Pay for Tax Collectors," *Quarterly Journal of Economics*, 2016, *131* (1), 219–271.

__ , __ , and __ , "Making Moves Matter: Experimental Evidence on Incentivizing Bureaucrats through Performance-Based Postings," *American Economic Review*, 2019, *109* (1), 237–270.

Kremer, Michael and Tom Wilkening, "Protecting Antiquities: A Role for Long-Term Leases?," *Journal of Economic Perspectives*, 2025, *39* (3), 127–148.

Lavery, Brian, *Nelson's Navy: The Ships, Men and Organisation, 1793–1815*, London: Conway Maritime Press, 1989.

Lewis, Michael, *A Social History of the Navy, 1793–1815*, London: George Allen & Unwin, 1960.

Longair, Sarah and John McAleer, "Introduction: Curating Empire: Museums and the British Imperial Experience," in Sarah Longair and John McAleer, eds., *Curating Empire: Museums and the British Imperial Experience*, Manchester: Manchester University Press, 2012, pp. 1–16.

MacDonald, Isobel, "Counting When, Who and How: Visualizing the British Museum's History of Acquisition through Collection Data, 1753–2019," *Journal of the History of Collections*, 2023, *35* (2), 305–320.

MacKenzie, John M., *Museums and Empire: Natural History, Human Cultures and Colonial Identities*, Manchester: Manchester University Press, 2009. Studies in Imperialism.

References IX

Miller, Charles and K. Shuvo Bakar, "Conflict Events Worldwide Since 1468BC: Introducing the Historical Conflict Event Dataset," *Journal of Conflict Resolution*, 2023, *67* (2–3), 522–554.

Mullahy, John and Edward C. Norton, "Why Transform Y? The Pitfalls of Transformed Regressions with a Mass at Zero," *Oxford Bulletin of Economics and Statistics*, 2024, *86* (2), 417–447.

Nunn, Nathan, "The Long-Term Effects of Africa's Slave Trades," *Quarterly Journal of Economics*, 2008, *123* (1), 139–176.

O'Brien, Patrick K., "The Costs and Benefits of British Imperialism 1846–1914," *Past & Present*, 1988, *120* (1), 163–200.

Offer, Avner, "The British Empire, 1870–1914: A Waste of Money?," *Economic History Review*, 1993, *46* (2), 215–238.

Oosterlinck, Kim, "Art as a Wartime Investment: Conspicuous Consumption and Discretion," *Economic Journal*, 2017, *127* (607), 2665–2701.

Parliamentary Papers, "Army—India—Kirwee Prize Money," Hansard, House of Commons, 3rd ser., vol. 201, cols. 1524–1551, 27 May 1870. `https://api.parliament.uk/historic-hansard/commons/1870/may/27/army-india-kirwee-prize-money`.

Parrillo, Nicholas R., *Against the Profit Motive: The Salary Revolution in American Government, 1780–1940*, New Haven: Yale University Press, 2013.

References X

Patrizio Gunning, Lucia and Debbie Challis, "Planned Plunder, the British Museum, and the 1868 Maqdala Expedition," *Historical Journal*, 2023, *66* (3), 550–572.

Querubín, Pablo and James M. Snyder Jr., "The Control of Politicians in Normal Times and Times of Crisis: Wealth Accumulation by U.S. Congressmen, 1850–1880," *Quarterly Journal of Political Science*, 2013, *8* (4), 409–450.

Ray, Debraj and Joan Esteban, "Conflict and Development," *Annual Review of Economics*, 2017, *9*, 263–293.

Redlich, Fritz, *De Praeda Militari: Looting and Booty, 1500–1815*, Wiesbaden: Franz Steiner Verlag, 1956. Vierteljahrschrift für Sozial- und Wirtschaftsgeschichte, Beiheft 39.

République française, "Loi No. 2026-351 du 9 mai 2026 relative à la restitution de biens culturels ayant fait l'objet d'une appropriation illicite," 2026. Journal officiel de la République française, 10 May 2026. `https://www.legifrance.gouv.fr/eli/loi/2026/5/9/2026-351/jo/texte`.

Sánchez de la Sierra, Raúl, "On the Origins of the State: Stationary Bandits and Taxation in Eastern Congo," *Journal of Political Economy*, 2020, *128* (1), 32–74.

Sandholtz, Wayne, *Prohibiting Plunder: How Norms Change*, New York: Oxford University Press, 2007.

Santos Silva, J. M. C. and Silvana Tenreyro, "The Log of Gravity," *Review of Economics and Statistics*, 2006, *88* (4), 641–658.

References XI

Sarr, Felwine and Bénédicte Savoy, "The Restitution of African Cultural Heritage: Toward a New Relational Ethics," Report to the French Ministry of Culture, 2018.

Savoy, Bénédicte, *Africa's Struggle for Its Art: History of a Postcolonial Defeat*, Princeton: Princeton University Press, 2022.

Silverman, Brian S. and Paul Ingram, "Asset Ownership and Incentives in Early Shareholder Capitalism: Liverpool Shipping in the Eighteenth Century," *Strategic Management Journal*, 2017, *38* (4), 854–875.

Smith, Adam, *An Inquiry into the Nature and Causes of the Wealth of Nations*, London: W. Strahan and T. Cadell, 1776.

Spiers, Edward M., *The Army and Society, 1815–1914*, London: Longman, 1980.

__ , "Spoils of War: Custom and Practice," in Henrietta Lidchi and Stuart Allan, eds., *Dividing the Spoils: Perspectives on Military Collections and the British Empire*, Manchester: Manchester University Press, 2020, pp. 19–38.

Stahn, Carsten, *Confronting Colonial Objects: Histories, Legalities, and Access to Culture*, Oxford: Oxford University Press, 2023.

Sun, Liyang and Sarah Abraham, "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects," *Journal of Econometrics*, 2021, *225* (2), 175–199.

Tilly, Charles, *Coercion, Capital, and European States, AD 990–1990*, Oxford: Basil Blackwell, 1990.

References XII

Tythacott, Louise, "Exhibiting and Auctioning Yuanmingyuan ('Summer Palace') Loot in 1860s and 1870s London: The Elgin and Negroni Collections," *Journal for Art Market Studies*, 2018b, *2* (3).

Voigtländer, Nico and Hans-Joachim Voth, "Gifts of Mars: Warfare and Europe's Early Rise to Riches," *Journal of Economic Perspectives*, 2013, *27* (4), 165–186.

Voth, Hans-Joachim and Guo Xu, "Discretion and Destruction: Promotions, Performance, and Patronage in the Royal Navy," 2022. Working paper, February 2022. An earlier version circulated as "Patronage for Productivity: Selection and Performance in the Age of Sail," CEPR Discussion Paper 13963, 2019.

Watt, Patrick, "'Ashantee Loot is Unique': British Military Culture and the Taking of Objects in the Third Anglo-Asante War, 1873–1874," *British Journal for Military History*, 2023, *9* (3), 21–40.

Williamson, Jeffrey G., "The Structure of Pay in Britain, 1710–1911," *Research in Economic History*, 1982, *7*, 1–54.

Wolseley, Garnet, *The Soldier's Pocket-Book for Field Service*, 5th ed., London: Macmillan, 1886.

Wood, Andrew B., "The Limits of Social Mobility: Social Origins and Career Patterns of British Generals, 1688–1815." PhD dissertation, London School of Economics 2011.

Xu, Guo, "The Costs of Patronage: Evidence from the British Empire," *American Economic Review*, 2018, *108* (11), 3170–3198.

Yule, Henry and Arthur Coke Burnell, *Hobson-Jobson: Being a Glossary of Anglo-Indian Colloquial Words and Phrases, and of Kindred Terms; Etymological, Historical, Geographical, and Discursive*, London: John Murray, 1886.